MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, KEY CODE ZJS

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic random access memory module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 7.693 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall ZJS.

3.5.2 Size. The module size shall be 2A.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 80 contact pins positioned 1A1 through 1A40 and 2A1 through 2A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/302-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, (V_{DD}) of 12.0 ±1.2 Vdc and (V_{BB}) of -5.0 ±0.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

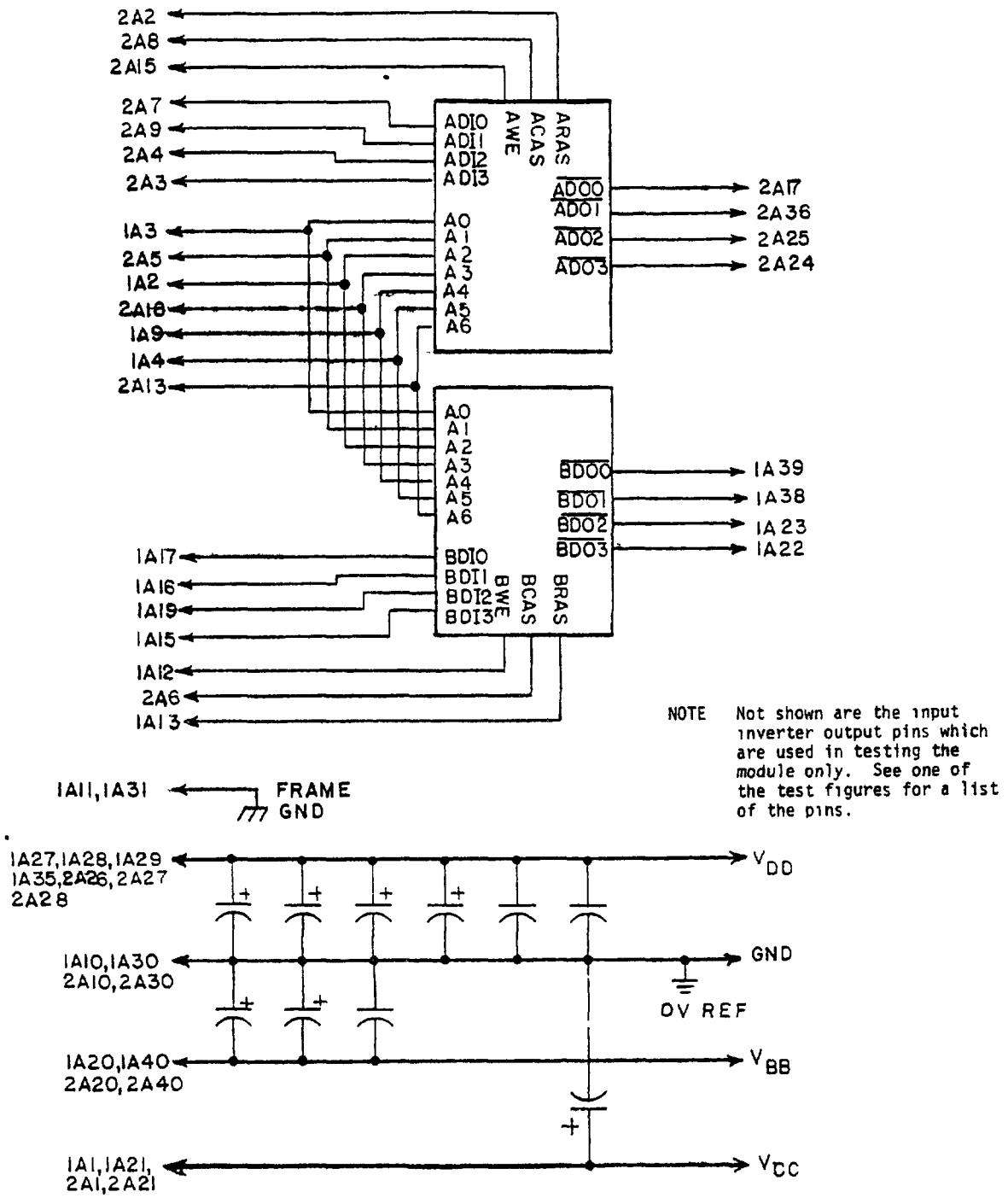
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE Not shown are the input inverter output pins which are used in testing the module only. See one of the test figures for a list of the pins.

FIGURE 1 ZJS module function diagram.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	II	S	10	-	10	-	10	-	10	-	M
A2-A10	Isolation, unused pins	II	X	10	-	10	-	10	-	10	-	M
B1-B5	Power supply stress check	III	S	← Must pass all remaining tests →								
C1-C21	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
D1-D21	Input clamp diode voltage	V	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E5	Output leakage current, low level	VI	X	-10	+10	-10	+10	-10	+10	-10	+10	µA
F1-F8	Output leakage current, high level	VI	X	-10	+10	-10	+10	-10	+10	-10	+10	µA
G1-G21	Input current, low level	VII	X	-0.01	-0.43	-0.01	-0.43	-0.01	-0.43	-0.01	-0.43	mA
H1-H21	Input current, high level	VII	X	0	20	0	20	0	20	0	20	µA
J1-J5	Ground pin continuity	VIII	X	0	1	0	1	0	1	0	1	
J6-J10	Ground pin continuity	VIII	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
K1-K8	Output voltage, low level (V_{OL})	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L8	Output voltage, high level (V_{OH})	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
M1	Power supply current (I_{DD})	XII	X	30	280	30	280	30	280	30	280	mA
M2,M3,M8	Power supply current (I_{CC})	XII	X	5	81	5	81	5	81	5	81	mA
M4,M5	Power supply current (I_{BB})	XII	X	0.1	3.2	0.1	3.2	0.1	3.2	0.1	3.2	mA
M6	Power supply current (I_{DD})	XII	X	20	216	20	216	20	216	20	216	mA
M7	Power supply current (I_{DD})	XII	X	1	18	1	18	1	18	1	18	mA
M9	Power supply current (I_{BB})	XII	X	0.1	1.6	0.1	1.6	0.1	1.6	0.1	1.6	µA
N1-N25	Dynamic functional test, C_L = 50 pF	XIII	X	← Must pass all dynamic functional tests →								
P1-P25	Dynamic functional test, C_L = 150 pF	XIII	Q	← Must pass all dynamic functional tests →								
Q1-Q21	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	X	3	30	3	30	3	32	3	32	ns
R1-R21	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	X	3	30	3	30	3	32	3	32	ns
S1-S6	Input crosstalk, low level	XV	Q	0	500	0	500	0	500	0	500	mV
T1-T6	Input crosstalk, high level	XV	Q	0	500	0	500	0	500	0	500	mV

See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/ 2/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J1-J3	Power supply transient amplitude	XVI	Q	0	500	0	500	0	500	0	500	mV
V1	Power supply filter capacitance (V_{DD})	XVII	S	13.2	24.6	13.2	24.6	13.2	24.6	13.2	24.6	µF
V2	Power supply filter capacitance (V_{CC})	XVII	S	1.54	2.86	1.54	2.86	1.54	2.86	1.54	2.86	µF
V3	Power supply filter capacitance (V_{BB})	XVII	S	3.12	5.78	3.12	5.78	3.12	5.78	3.12	5.78	µF
W1-W21	Input capacitance	XVII	Q	2	15	2	15	2	15	2	15	pF
X1-X8	Output capacitance	XVII	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification inspection. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

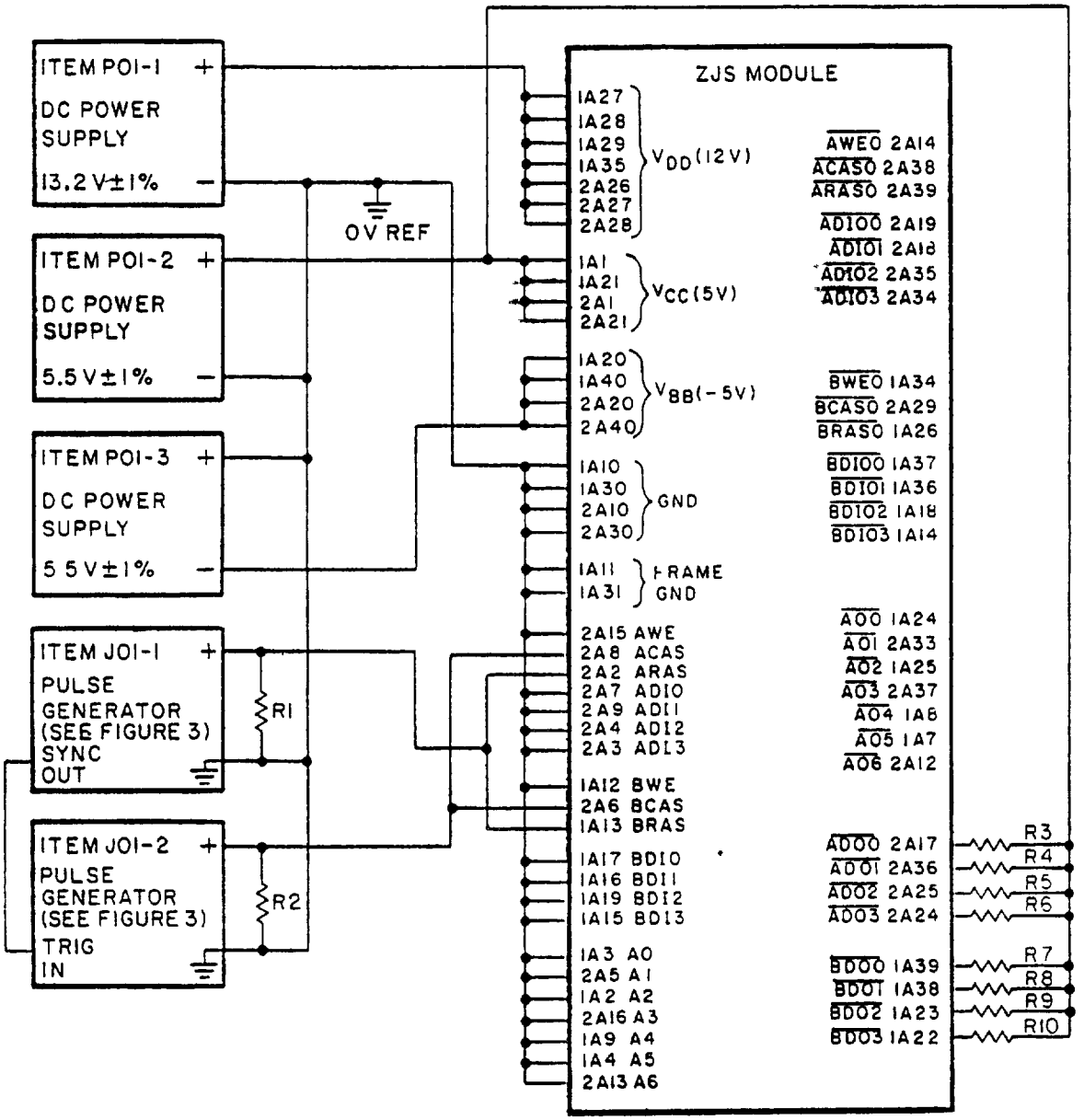
6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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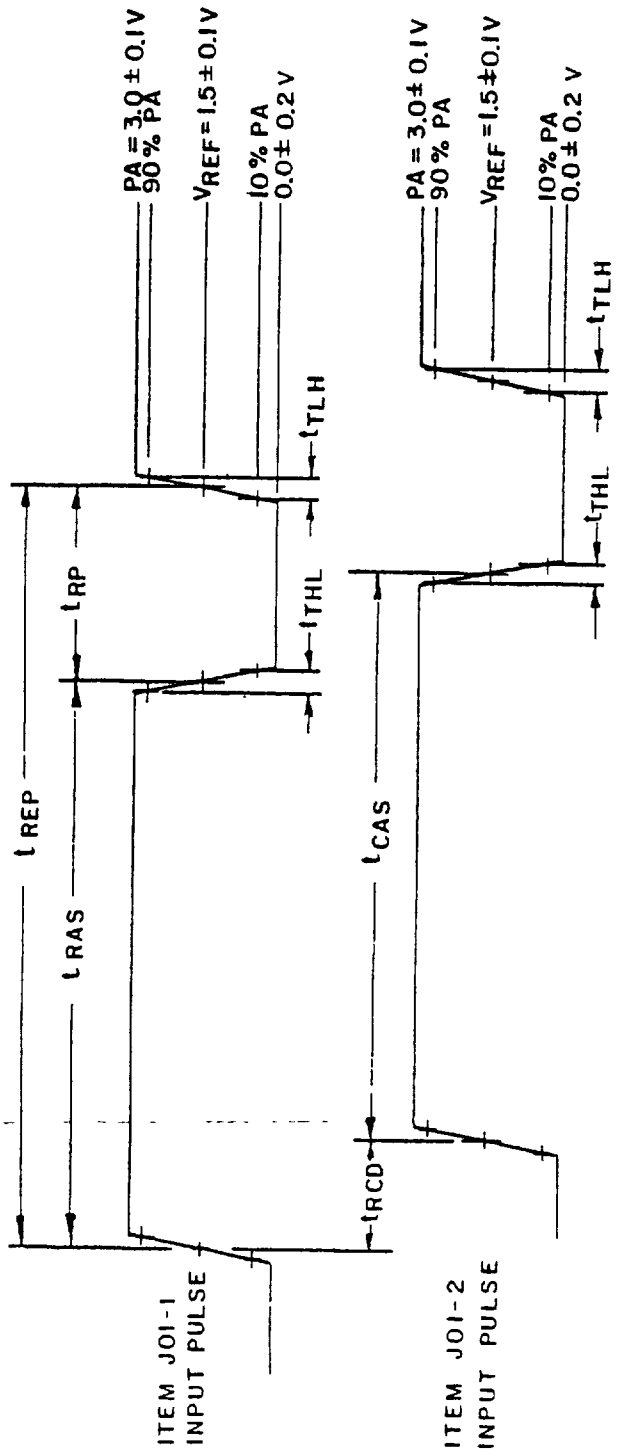
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W
2 R3-R10 = 1.21 kΩ ±1%, 1/4 W

FIGURE 2 Operating temperature, life and component temperature test circuit

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NOTE Input parameters
t_{RAS} = 220 ±5 ns
t_{RP} = 180 ±5 ns
t_{RCD} = 65 ±5 ns
t_{CAS} = 155 ±5 ns
t_{REP} = 400 ±5 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns

FIGURE 3 Operating temperature, life and component temperature test circuit waveforms for figure 2.

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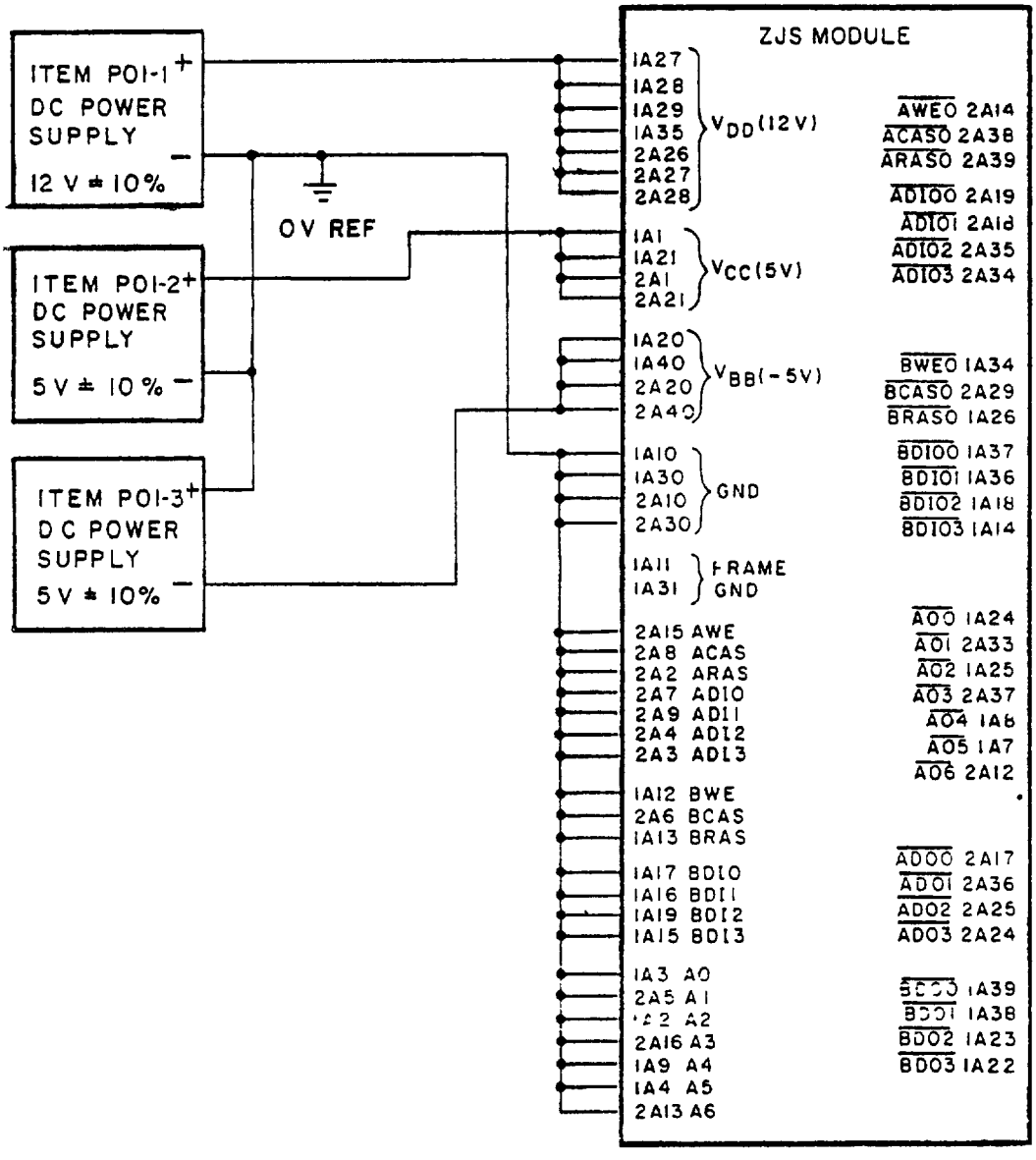


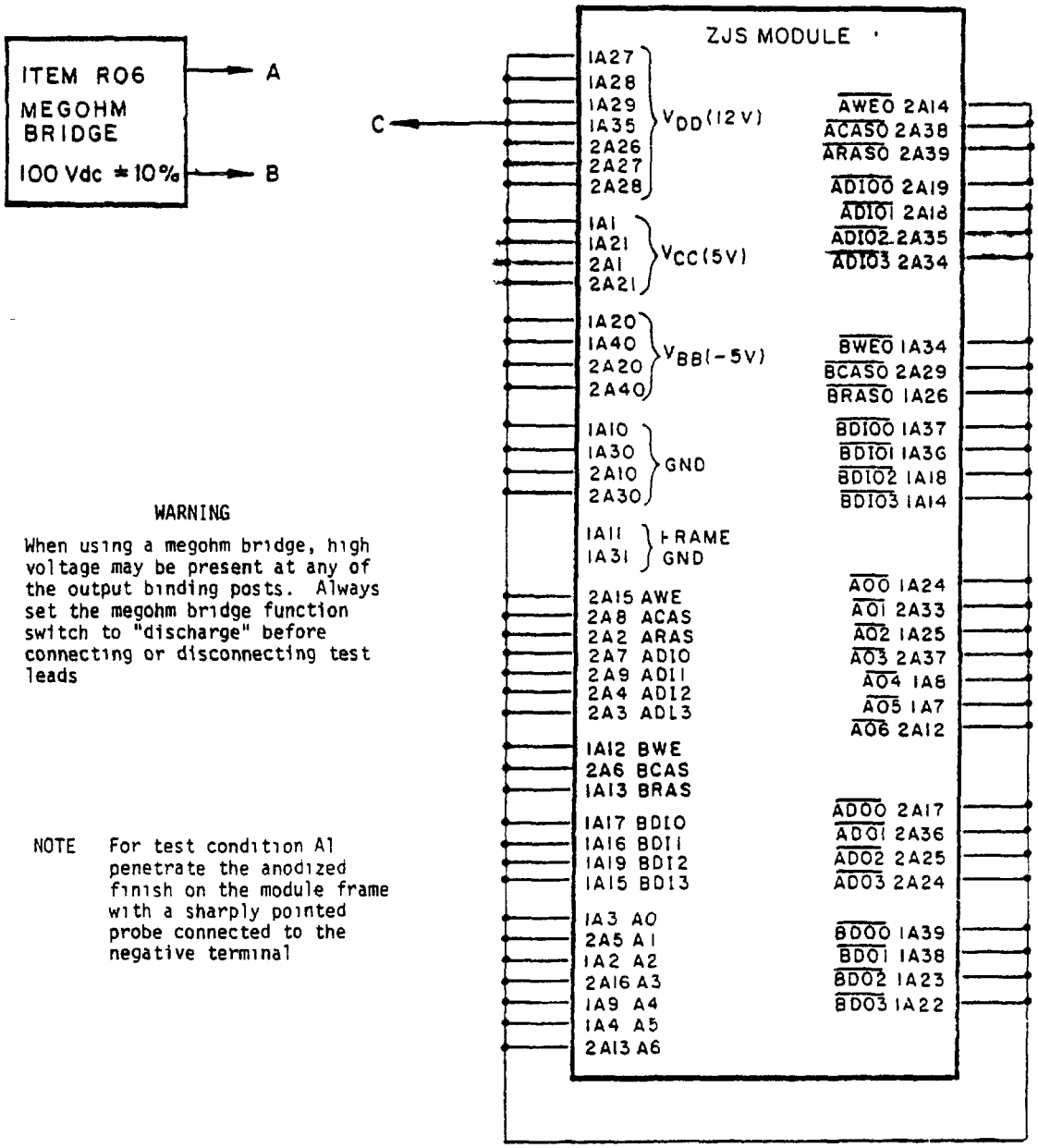
FIGURE 4. Vibration test circuit

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C, 1A5, 1A6, 1A32, 1A33, 2A11, 2A23, 2A22, 2A31, 2A32	frame	10	—	MΩ	5
A2	Isolation, unused pins	C, 1A6, 1A32, 1A33, 2A11, 2A22, 2A23, 2A31, 2A32	1A5				
A3		C, 1A5, 1A32, 1A33, 2A11, 2A22,2A23,2A31,2A32	1A6				
A4		C, 1A5, 1A6, 1A33, 2A11, 2A23, 2A22, 2A31, 2A32	1A32				
A5		C, 1A5, 1A6, 1A32, 2A11, 2A23, 2A22, 2A31, 2A32	1A33				
A6		C, 1A5, 1A6, 1A32, 1A33, 2A23, 2A22, 2A31, 2A32	2A11				
A7		C, 1A5, 1A6, 1A32, 1A33, 2A11, 2A22, 2A31, 2A32	2A23				
A8		C, 1A5, 1A6, 1A32, 1A33, 2A11, 2A23, 2A31, 2A32	2A22				
A9		C, 1A5, 1A6, 1A32, 1A33, 2A11, 2A23, 2A22, 2A32	2A31				
A10		C, 1A5, 1A6, 1A32, 1A33, 2A11, 2A23, 2A22, 2A31	2A32				

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting test leads

NOTE For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 5 Isolation test circuit for table II

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TABLE III. Power supply stress check test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Set in order shown	Test figure
B1	V_{BB} to V_{DD} Power supply stress check	E	1A20,1A40, 2A20,2A40	1A1,1A21, 2A1,2A21	1A27,1A28, 1A29,1A35, 2A26,2A27, 2A28	A	Open	P01-1 to 8.0 V +0, -1% P01-2 to 5.0 V ±1% P01-3 to 12.0 V +0, -1%	6
B2	V_{CC}	E	1A20,1A40, 2A20,2A40	1A1,1A21, 2A1,2A21	1A27,1A28, 1A29,1A35, 2A26,2A27, 2A28	A	Open	P01-1 to 5.0 V ±1% P01-2 to 7.0 V +0, -1% P01-3 to 12.0 V ±1%	
B3	V_{DD}	E	1A20,1A40, 2A20,2A40	1A1,1A21, 2A1,2A21	1A27,1A28, 1A29,1A35, 2A26,2A27, 2A28	A	Open	P01-1 to 5.0 V +0, -1% P01-2 to 5.0 V ±1% P01-3 to 15.0 V +0, -1%	
B4	V_{BB} to input	E	1A20,1A40, 2A20,2A40	Open	Open	1A1,1A21, 1A27,1A28, 1A29,1A35, 2A1,2A21, 2A26,2A27, 2A28,A	1/	P01-1 to 20 V +0, -1% P01-4 to 3.0 V ±1%	
B5	Repeat V_{BB} to input	1A20, 1A40, 2A20, 2A40	E	Open	Open	1A1,1A21, 1A27,1A28, 1A29,1A35, 2A1,2A21, 2A26,2A27, 2A28,B	1/	P01-1 to 0.5 V +0, -1% P01-4 to 3.0 V ±1%	

1/ For test conditions B4 and B5 connect point E to pins 1A2, 1A3, 1A4, 1A9, 1A12, 1A13, 1A15, 1A16, 1A17, 1A19, 2A2, 2A3, 2A4, 2A5, 2A6, 2A7, 2A8, 2A9, 2A13, 2A15, 2A16

TABLE IV. Input breakdown current test

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
C1	ABE	Input breakdown current	1A2,1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A13,2A16	2A15	0	100	µA	7
C2	ABAS	Input breakdown current	1A2,1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A7,2A9,2A13, 2A15,2A16	2A8	0	100	µA	7
C3	AKAS	Input breakdown current	1A2,1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A3- 2A9,2A13,2A15, 2A16	2A2	0	100	µA	7

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TABLE IV Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C4	ADIO Input breakdown current	1A2,1A3,1A4,1A9, 1A12,1A13,1A15-1A17,1A19,2A2-2A6,2A8,2A9, 2A13,2A15,2A16	2A7	0	100	µA	7
C5	ADI1	1A2,1A3,1A4,1A9, 1A12,1A13,1A15-1A17,1A19,2A2-2A8,2A13,2A15, 2A16	2A9				
C6	ADI2	1A2,1A3,1A4,1A9, 1A12,1A13,1A15-1A17,1A19,2A2, 2A3,2A5-2A9, 2A13,2A15,2A16	2A4				
C7	ADI3	1A2,1A3,1A4,1A9, 1A12,1A13,1A15-1A17,1A19,2A2, 2A4-2A9,2A13, 2A15,2A16	2A3				
C8	BWE	1A2,1A3,1A4,1A9, 1A13,1A15-1A17, 1A19,2A2-2A9, 2A13,2A15,2A16	1A12				
C9	BCAS	1A2,1A3,1A4,1A9, 1A12,1A13,1A15-1A17,1A19,2A2-2A5,2A7-2A9, 2A13,2A15,2A16	2A6				
C10	BRAS	1A2,1A3,1A4,1A9, 1A12,1A15-1A17, 1A19,2A2-2A9, 2A13,2A15,2A16	1A13				
C11	BDIO	1A2,1A3,1A4,1A9, 1A12,1A15,1A16, 1A19,1A13,2A2-2A9,2A13,2A15, 2A16	1A17				
C12	BDI1	1A2,1A3,1A4,1A9, 1A12,1A15,1A17, 1A19,1A13,2A2-2A9,2A13,2A15, 2A16	1A16				
C13	BDI2	1A2,1A3,1A4,1A9, 1A12,1A15-1A17, 1A13,2A2-2A9, 2A13,2A15,2A16	1A19				

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TABLE IV Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C14	BDI3	Input breakdown current	1A2,1A3,1A4,1A9, 1A12,1A16,1A17, 1A19,1A13,2A2- 2A9,2A13,2A15, 2A16	1A15	0	100	MA	7
C15	A0		1A2,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A13,2A15, 2A16	1A3				
C16	A1		1A2,1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A4,2A6-2A9, 2A13,2A15,2A16	2A5				
C17	A2		1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A13,2A15, 2A16	1A2				
C18	A3		1A2,1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A13,2A15	2A16				
C19	A4		1A2,1A3,1A4, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A13,2A15, 2A16	1A9				
C20	A5		1A2,1A3,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A13,2A15, 2A16	1A4				
C21	A6		1A2,1A3,1A4,1A9, 1A12,1A13,1A15- 1A17,1A19,2A2- 2A9,2A15,2A16	2A13				

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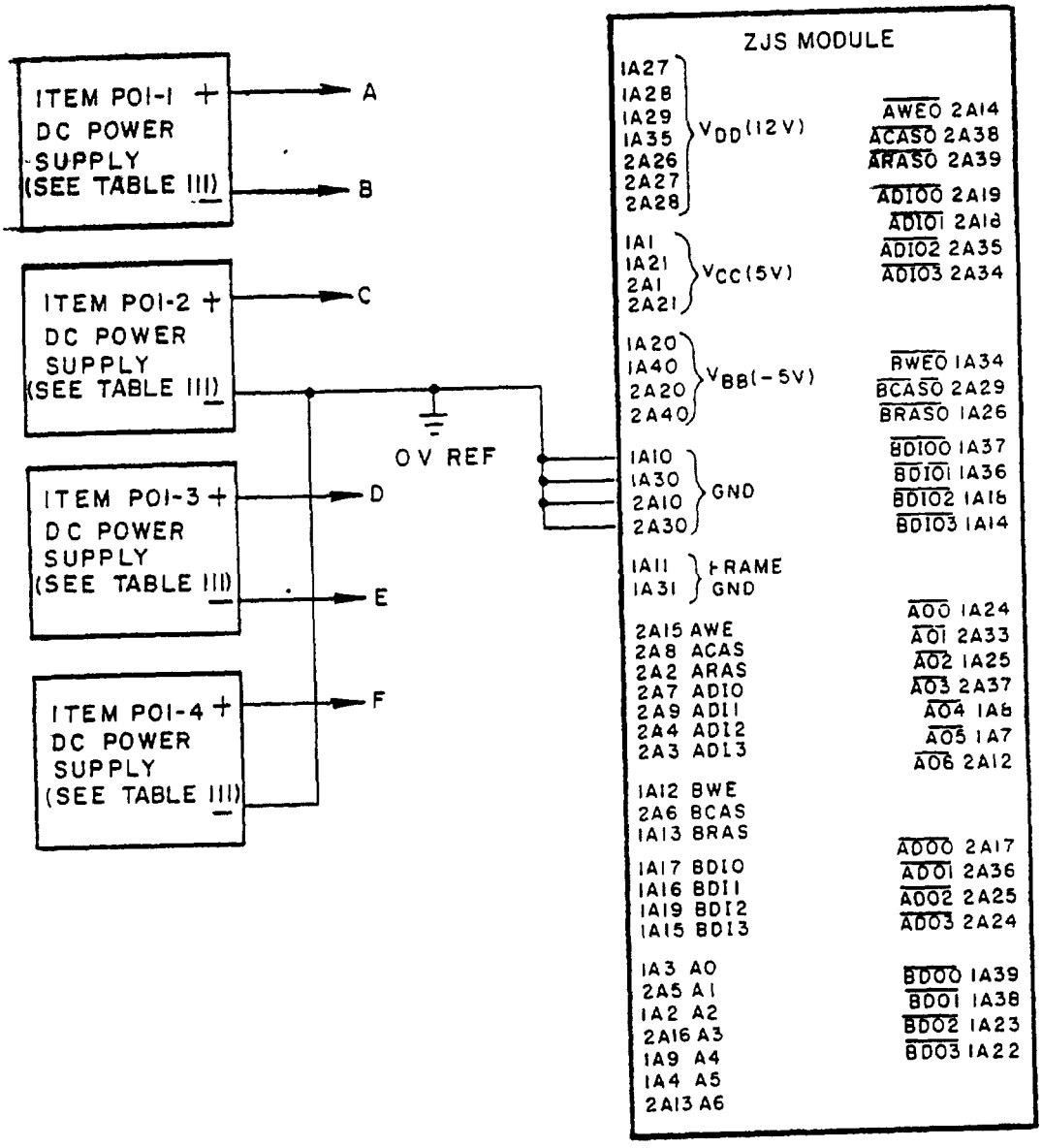


FIGURE 6 Power supply stress check test circuit for table III.

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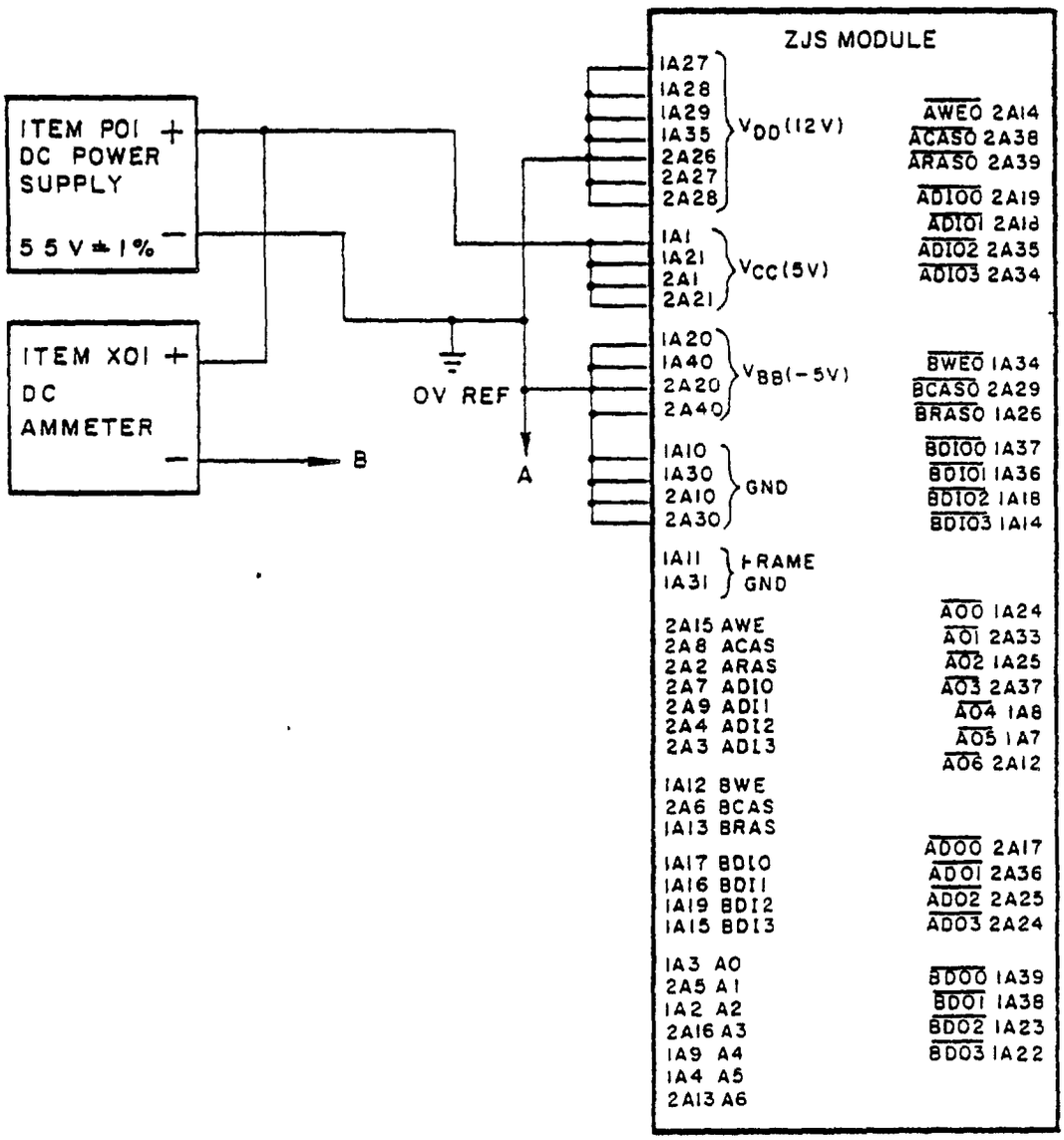


FIGURE 7 Input breakdown current test circuit for table IV

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TABLE V Input clamp diode voltage test

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
D1	AWE Input clamp diode voltage	2A15	-0.4	-1.5	V	8
D2	ACAS	2A8				
D3	ARAS	2A2				
D4	ADI0	2A7				
D5	ADI1	2A9				
D6	ADI2	2A4				
D7	ADI3	2A3				
D8	BWE	1A12				
D9	BCAS	2A6				
D10	BRAS	1A13				
D11	BDI0	1A17				
D12	BDI1	1A16				
D13	BDI2	1A19				
D14	BDI3	1A15				
D15	A0	1A3				
D16	A1	2A5				
D17	A2	1A2				
D18	A3	2A16				
D19	A4	1A9				
D20	A5	1A4				
D21	A6	2A13				

TABLE VI Output leakage current (low-level and high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
F1	$\overline{ADO0}$ Output leakage current, low level	2A17	-10	+10	µA	9
F2	$\overline{ADO1}$	2A36				
F3	$\overline{ADO2}$	2A25				
F4	$\overline{ADO3}$	2A24				

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TABLE VI. Output leakage current (low-level and high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
E5	$\overline{BD00}$	Output leakage current, low level	1A39	-10	+10	uA	9
E6	$\overline{BD01}$		1A38				
E7	$\overline{BD02}$		1A23				
E8	$\overline{BD03}$		1A22				
F1	$\overline{AD00}$	Output leakage current, high level	2A17				
F2	$\overline{AD01}$		2A36				
F3	$\overline{AD02}$		2A25				
F4	$\overline{AD03}$		2A24				
F5	$\overline{BD00}$		1A39				
F6	$\overline{BD01}$		1A38				
F7	$\overline{BD02}$		1A23				
F8	$\overline{BD03}$		1A22				

TABLE VII Input current (low-level and high-level) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
G1	AWE	Input current, low level	Open	Open	2A15	-0.01	-0.43	mA	10
G2	ACAS				2A8				
G3	ARAS				2A2				
G4	AD10				2A7				
G5	AD11				2A9				
G6	AD12				2A4				
G7	AD13				2A3				
G8	BWE				1A12				
G9	BCAS				2A6				
G10	BRAS				1A13				
G11	BD10				1A17				

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TABLE VII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G12	BDI1	Input current, low level	Open	Open	1A16	-0.01	-0.43	mA	10
G13	BDI2				1A19				
G14	BDI3				1A15				
G15	A0				1A3				
G16	A1				2A5				
G17	A2				1A2				
G18	A3				2A16				
G19	A4				1A9				
G20	A5				1A4				
G21	A6				2A13				
H1	AWE	Input current, high level	2A15	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A6,2A7,2A8,2A9,2A13, 2A16	Open	0	20	µA	
H2	ACAS		2A8	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A6,2A7,2A9,2A13,2A15, 2A16					
H3	ARAS		2A2	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H4	ADI0		2A7	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A6,2A8,2A9,2A13,2A15, 2A16					
H5	ADI1		2A9	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A6,2A7,2A8,2A13,2A15, 2A16					
H6	ADI2		2A4	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H7	ADI3		2A3	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					

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TABLE VII Input current (low-level and high-level) test - Continued

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
H8	BWE	Input current, high level	1A12	1A2,1A3,1A4,1A9,1A13, 1A15,1A16,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16	Open	0	20	uA	10
H9	BCAS		2A6	1A2,1A3,1A4,1A9,1A12, 1A13 1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A7,2A8,2A9,2A13,2A15, 2A16					
H10	BRAS		1A13	1A2,1A3,1A4,1A9,1A12, 1A15,1A16,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H11	BDI0		1A17	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H12	BDI1		1A16	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H13	BDI2		1A19	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H14	BDI3		1A15	1A2,1A3,1A4,1A9,1A12, 1A13,1A16,1A17,1A19, 2A2,2A3 2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H15	A0		1A3	1A2,1A4,1A9,1A12,1A13 1A15,1A16,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H16	A1		2A5	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					

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TABLE VII Input current (low-level and high-level) test - Continued

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H17	A2	Input current, high level	1A2	1A3,1A4,1A9,1A12,1A13, 1A15,1A16,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16	Open	0	20	µA	10
H18	A3		2A16	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A6,2A7,2A8,2A9,2A13, 2A15					
H19	A4		1A9	1A2,1A3,1A4,1A12,1A13, 1A15,1A16,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H20	A5		1A4	1A2,1A3,1A9,1A12,1A13, 1A15,1A16,1A17,1A19, 2A2,2A3,2A4,2A5,2A6, 2A7,2A8,2A9,2A13,2A15, 2A16					
H21	A6		2A13	1A2,1A3,1A4,1A9,1A12, 1A13,1A15,1A16,1A17, 1A19,2A2,2A3,2A4,2A5, 2A6,2A7,2A8,2A9,2A15, 2A16					

TABLE VIII Ground pin continuity test

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	1A30	Ground pin continuity	1A30	1A10	0	1	Ω	11
J2	2A10		2A10	1A10				
J3	2A30		2A30	1A10				
J4	Frame		1A11	1A31				
J5	Frame to fin		1A11,1A31	Fin				
J6	1A30		1A30	1A10	0.0	0.1		
J7	2A10		2A10	1A10				
J8	2A30		2A30	1A10				
J9	Frame		1A11	1A31				
J10	Frame to fin		1A11,1A31	Fin				

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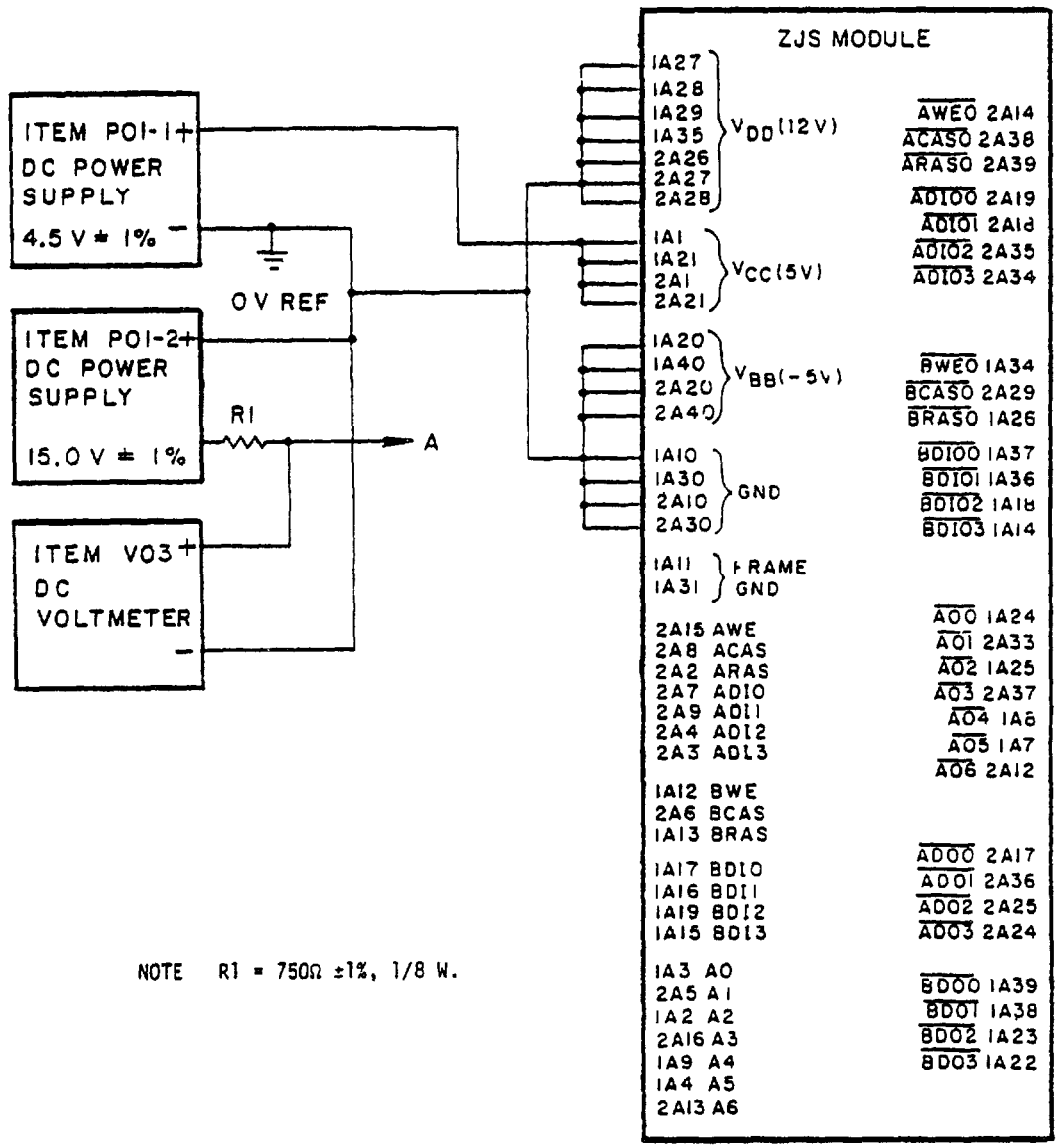
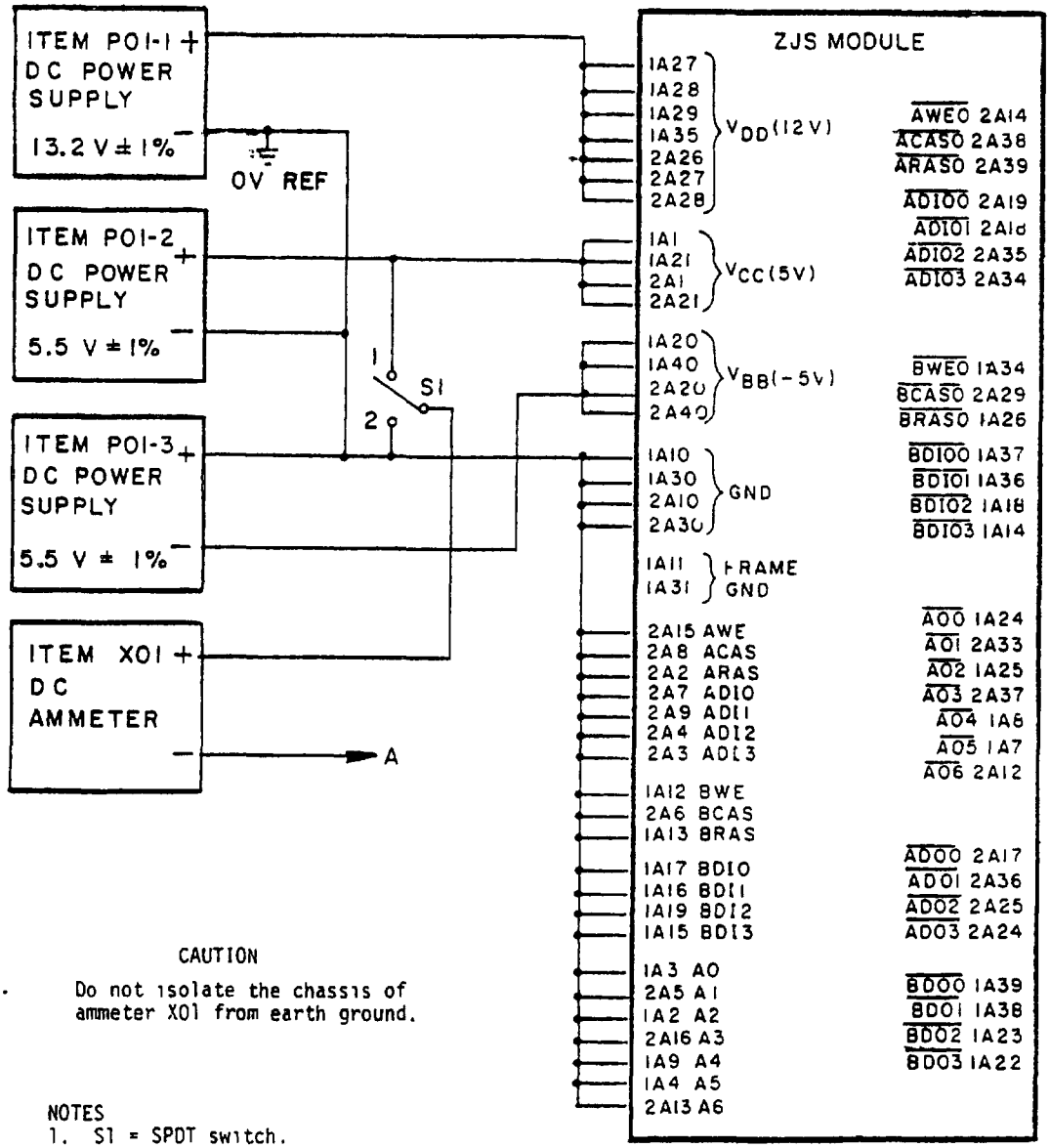


FIGURE 8. Input clamp diode voltage test circuit for table V

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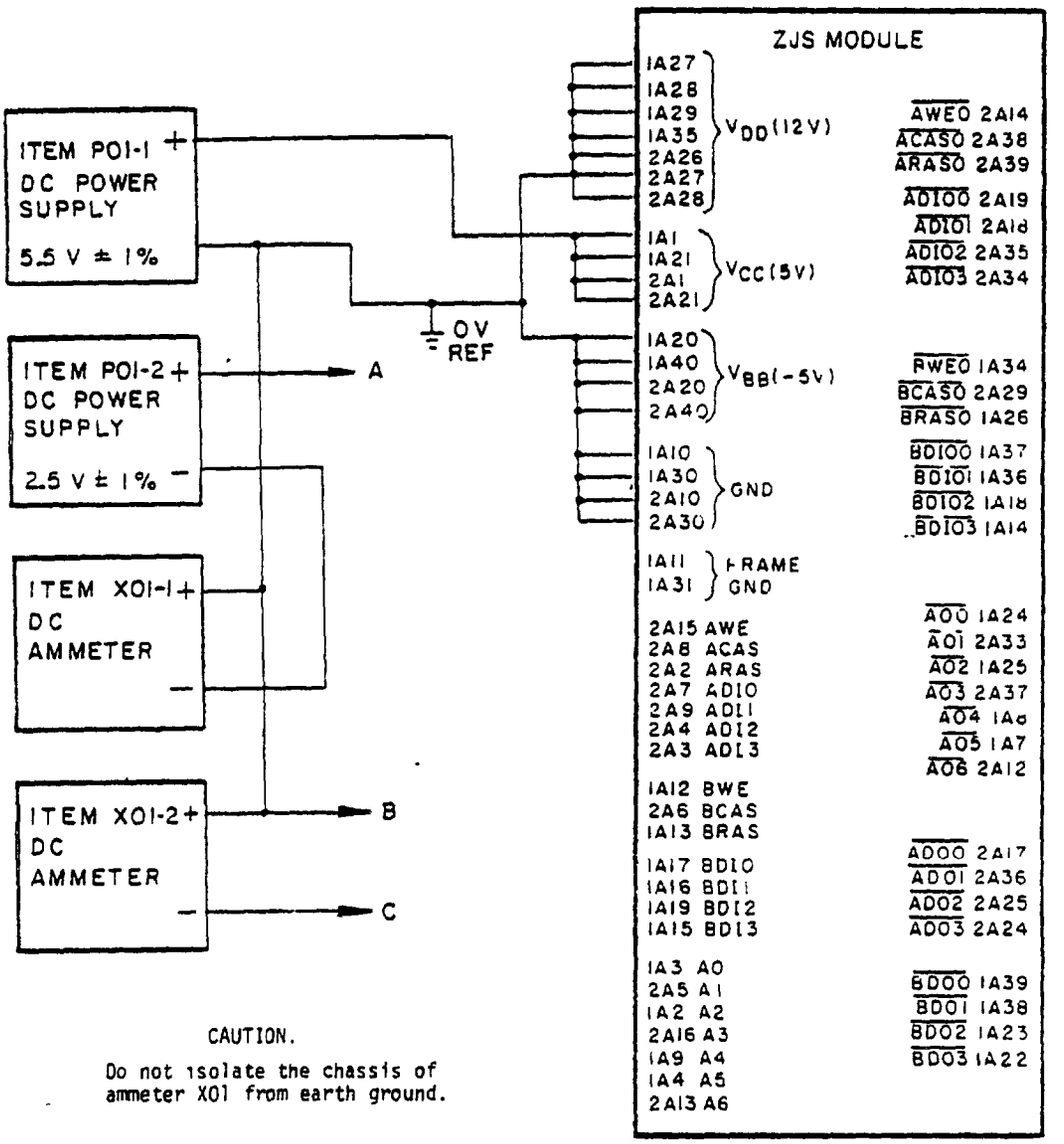
CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES
1. S1 = SPDT switch.
2. Switch S1 into position 2 for test conditions E1-E8.
3 Switch S1 into position 1 for test conditions F1-F8

FIGURE 9. Output leakage current (low-level and high-level) test circuit for table VI

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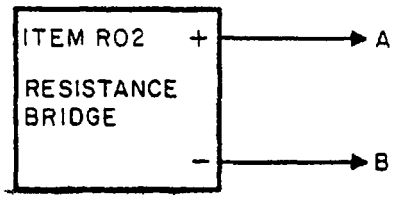
CAUTION.

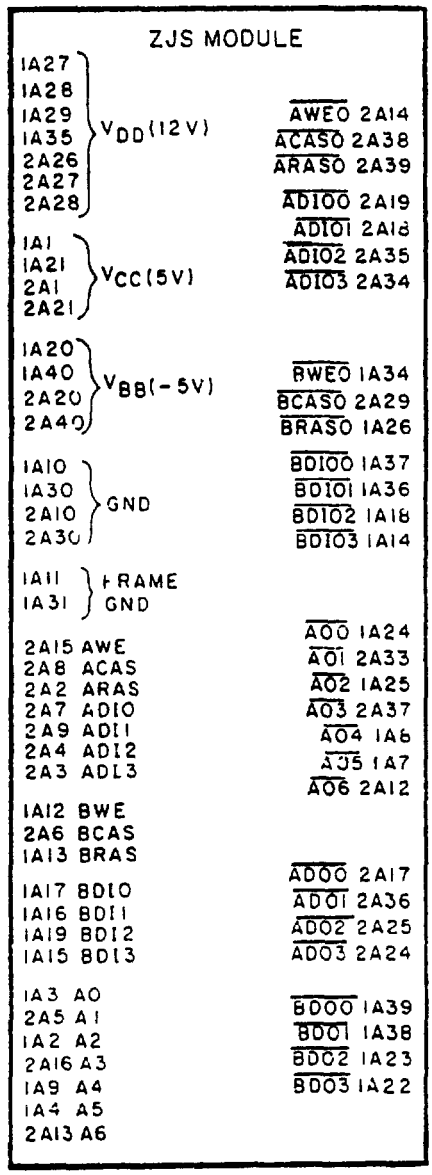
Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES
1. Since the ammeters are not used simultaneously, only one ammeter (XO1) is required
2. With point "A" disconnected, ammeter XO1-1 should indicate zero current on the 1.0 μA range

FIGURE 10 Input current (low-level and high-level) test circuit for table VII

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NOTE: For test conditions J5-J10, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 11 Ground pin continuity test circuit for table VIII

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TABLE IX Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
K1	$\overline{ADO0}$	Output voltage, low level (V_{OL})	2A17	0.0	0.4	V	12
K2	$\overline{ADO1}$		2A36				
K3	$\overline{ADO2}$		2A25				
K4	$\overline{ADO3}$		2A24				
K5	$\overline{BDO0}$		1A39				
K6	$\overline{BDO1}$		1A38				
K7	$\overline{BDO2}$		1A23				
K8	$\overline{BDO3}$		1A22				
L1	$\overline{ADO0}$	Output voltage, high level (V_{OH})	2A17	2.4	4.5		
L2	$\overline{ADO1}$		2A36				
L3	$\overline{ADO2}$		2A25				
L4	$\overline{ADO3}$		2A24				
L5	$\overline{BDO0}$		1A39				
L6	$\overline{BDO1}$		1A38				
L7	$\overline{BDO2}$		1A23				
L8	$\overline{BDO3}$		1A22				

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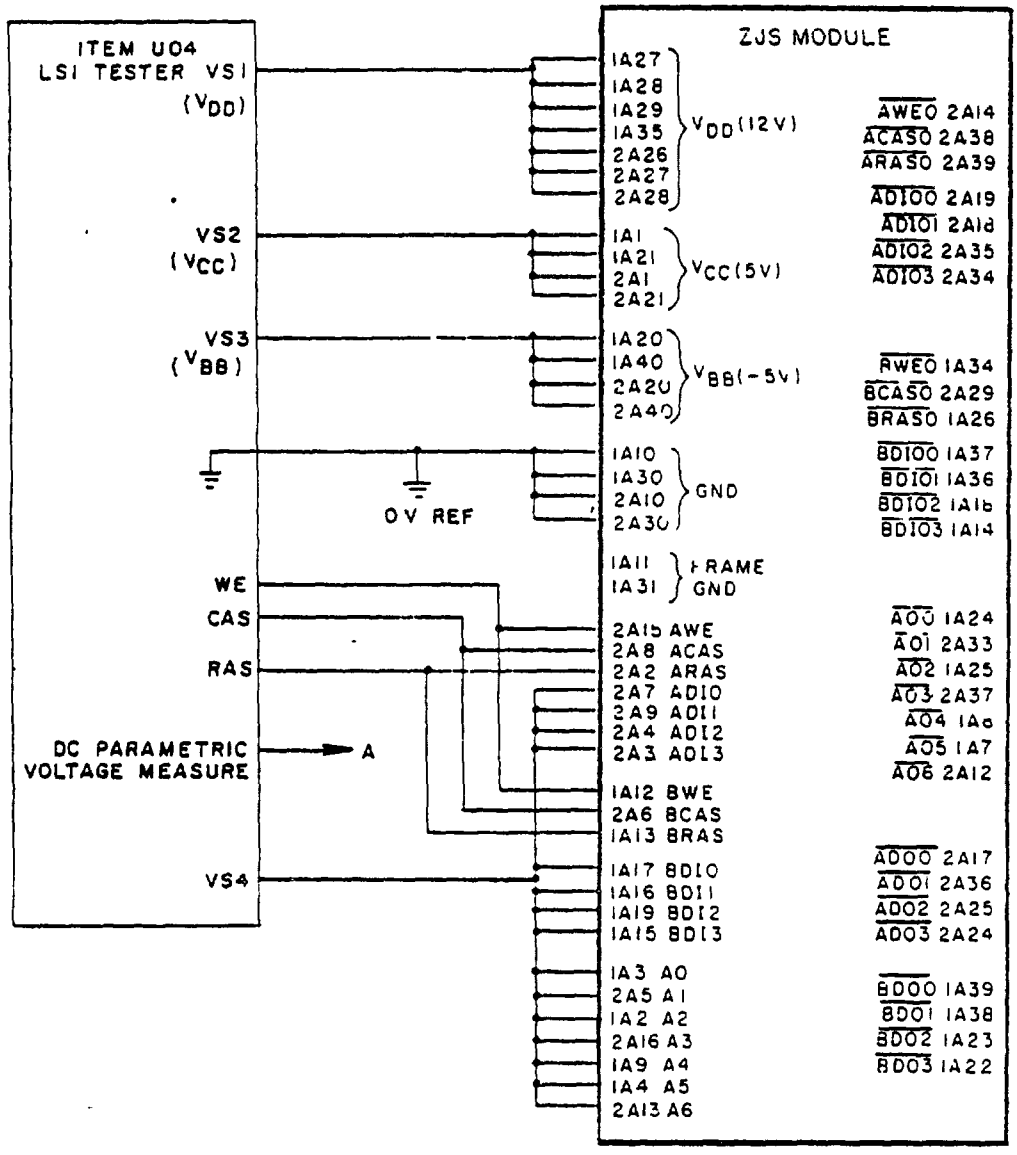
TABLE X. Test conditions for the output voltage (low-level) test in table IX.

Parameter	Input test conditions	Test figure
VS1 (V_{DD})	10.8 V ±1%	12, 13
VS2 (V_{CC})	4.5 V ±1%	
VS3 (V_{BB})	-4.5 V ±1%	
VS4 (address and data inputs)	3.0 ±0.2 V	
t_{RAS}	345 ±5 ns	
t_{RP}	155 ±2 ns	
t_{CAS}	300 ±4 ns	
t_{RWD}	180 ±3 ns	
t_{WP}	90 ±2 ns	
t_{TLH}, t_{THL}, (all inputs)	10 ±1 ns	

TABLE XI. Test conditions for the output voltage (high-level) test in table IX.

Parameter	Input test conditions	Test figure
VS1 (V_{DD})	10.8 V ±1%	12, 13
VS2 (V_{CC})	4.5 V ±1%	
VS3 (V_{BB})	-4.5 V ±1%	
VS4 (address and data inputs)	0.0 ±0.2 V	
t_{RAS}	345 ±5 ns	
t_{RP}	155 ±2 ns	
t_{CAS}	300 ±4 ns	
t_{RWD}	180 ±3 ns	
t_{WP}	90 ±2 ns	
t_{TLH}, t_{THL}, (all inputs)	10 ±1 ns	

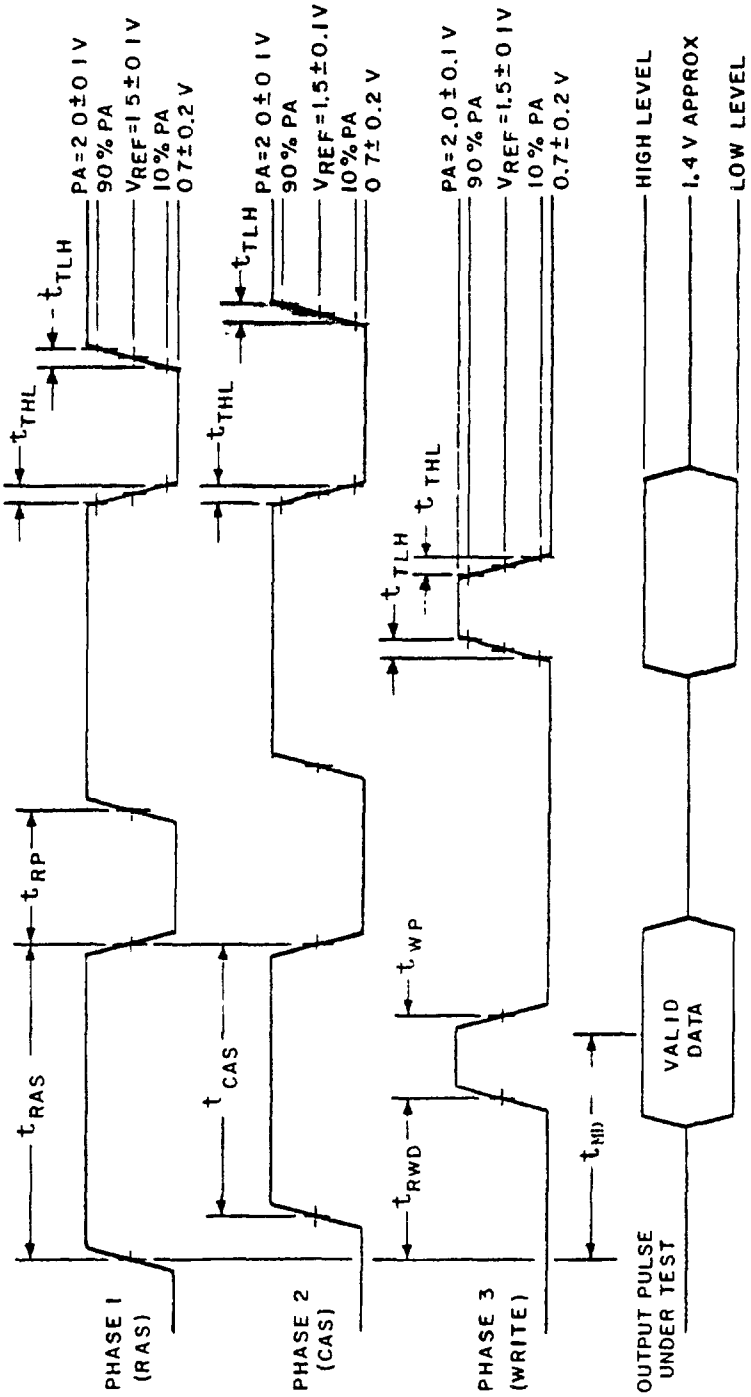
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NOTES
1. All tester leads to be of minimum length
2. Input conditions are found in table X for test conditions K1-K8 and table XI for test conditions L1-L8.
3. Load outputs per figure 14.

FIGURE 12. Output voltage (low-level and high-level) test circuit for table IX

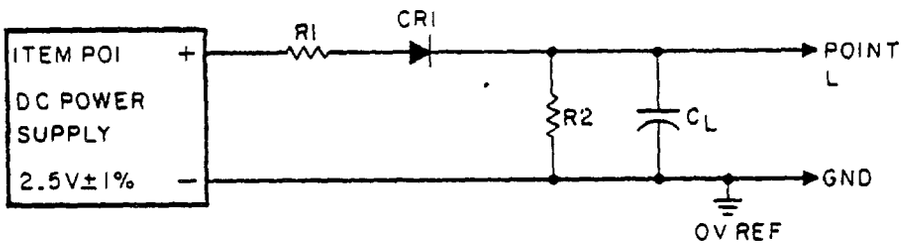
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NOTES
1 t_{MD} = Measurement delay = 225 ±10 ns
2 Allow a minimum 10 RAS cycles to occur before making output voltage measurements.

FIGURE 13. Output voltage (low-level and high-level) test circuit waveforms for table IX.

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NOTES.
1. CR1 = 1N3600, 1N4150 or equivalent
2. R2 = 475Ω ±1%, 1/8 W
3. R1 = Select a value for a current flow of 4.2 mA 5% out of point L when point L is held at 0.4 V
4. C_L = 50 ±5 pF (including probe and fixture capacitance) for all S tests
5. C_L = 150 ±10 pF (including probe and fixture capacitance) for all Q tests.
6. C_L is not required in test conditions K1-K8 and L1-L8

FIGURE 14 ZJS logic load circuit

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TABLE XII Power supply current test

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
M1	RAS/CAS cycling (I_{DD})	Power supply current	2A6,2A8	Open	1A13,2A2	30	280	mA	15,16
M2	RAS/CAS cycling (I_{CC})		2A6,2A8		1A13,2A2	5	81		15,16
M3	RAS/CAS cycling (I_{BB})		2A6,2A8		1A13,2A2	0.1	3.2		15,16
M4	Refresh cycling (I_{DD})		1A13,2A2		Open	20	216		15,17
M5	Refresh cycling (I_{CC})		1A13,2A2			5	81		15,17
M6	Refresh cycling (I_{BB})		1A13,2A2			0.1	3.2		15,17
M7	Standby (I_{DD})		Open	1A13,2A2, 2A6,2A8		1	18		15
M8	Standby (I_{CC})		Open	1A13,2A2, 2A6,2A8		5	81		15
M9	Standby (I_{BB})		Open	1A13,2A2, 2A6,2A8		0.1	1.6		15

TABLE XIII. Dynamic functional test

Test condition	Parameter	Set item U04 voltages to ±12 V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Test figure
N1	Checker board 0 test pattern, C_L = 50 pf	4.5	10.8	-4.5	Module shall conform to figure 25 requirements		18,19,20, 23,24
N2		4.5	10.8	-5.5			
N3		4.5	13.2	-4.5			
N4		4.5	13.2	-5.5			
N5		5.5	10.8	-4.5			
N6		5.5	10.8	-5.5			
N7		5.5	13.2	-4.5			
N8		5.5	13.2	-5.5			
N9	Checker board 1 test pattern C_L = 50 pf	4.5	10.8	-4.5			
N10		4.5	10.8	-5.5			
N11		4.5	13.2	-4.5			
N12		4.5	13.2	-5.5			

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TABLE XIII. Dynamic functional test - Continued

Test condition	Parameter	Set item U04 voltages to ±1%			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
N13	Checker board 1 test pattern, C_L = 50 pF	5.5	10.8	-4.5	Module shall conform to figure 25 requirements		18,19,20, 23,24
N14		5.5	10.8	-5.5			
N15		5.5	13.2	-4.5			
N16		5.5	13.2	-5.5			
N17	V bump test, C_L = 50 pF	See figure 26			Module shall conform to figure 26 requirements		
N18	Marching 1 test pattern, C_L = 50 pF	4.5	10.8	-4.5	Module shall conform to figure 27 requirements		18,19,21, 23,24
N19		4.5	10.8	-5.5			
N20		4.5	13.2	-4.5			
N21		4.5	13.2	-5.5			
N22		5.5	10.8	-4.5			
N23		5.5	10.8	-5.5			
N24		5.5	13.2	-4.5			
N25		5.5	13.2	-5.5			
N26	Marching 0 test pattern, C_L = 50 pF	4.5	10.8	-4.5			
N27		4.5	10.8	-5.5			
N28		4.5	13.2	-4.5			
N29		4.5	13.2	-5.5			
N30		5.5	10.8	-4.5			
N31		5.5	10.8	-5.5			
N32		5.5	13.2	-4.5			
N33		5.5	13.2	-5.5			
N34	Address complement 1 test, C_L = 50 pF	4.5	10.8	-4.5	Module shall conform to figure 28 requirements		18,19,20, 23,24
N35		4.5	10.8	-5.5			
N36		4.5	13.2	-4.5			
N37		4.5	13.2	-5.5			
N38		5.5	10.8	-4.5			
N39		5.5	10.8	-5.5			

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TABLE XIII Dynamic functional test - Continued.

Test condition	Parameter	Set item U04 voltages to ±1%			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
N40	Address complement 1 test, C_L = 50 pF	5 5	13 2	-4.5	Module shall conform to figure 28 requirements		18,19,20, 23,24
N41	Address complement 1 test, C_L = 50 pF	5.5	13 2	-5.5			
N42	Address complement 0 test, C_L = 50 pF	4 5	10 8	-4.5			
N43		4.5	10.8	-5 5			
N44		4.5	13 2	-4 5			
N45		4.5	13 2	-5 5			
N46		5.5	10 8	-4.5			
N47		5.5	10.8	-5 5			
N48		5 5	13 2	-4 5			
N49		5.5	13 2	-5 5			
N50	Walking diagonal 1 test pattern, C_L = 50 pF	4 5	10.8	-4.5	Module shall conform to figure 29 requirements		
N51		4.5	10 8	-5 5			
N52		4.5	13.2	-4 5			
N53		4.5	13.2	-5.5			
N54		5.5	10 8	-4 5			
N55		5.5	10 8	-5 5			
N56		5.5	13.2	-4.5			
N57		5.5	13 2	-5.5			
N58	Walking diagonal 0 test pattern, C_L = 50 pF	4.5	10 8	-4.5			
N59		4.5	10 8	-5 5			
N60		4.5	13 2	-4.5			
N61		4.5	13 2	-5.5			
N62		5.5	10.8	-4.5			
N63		5.5	10 8	-5 5			
N64		5 5	13 2	-4.5			
N65		5 5	13 2	-5 5			

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TABLE XIII. Dynamic functional test - Continued.

Test condition	Parameter	Set item U04 voltages to ±12			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
N66	Galloping rows 1 test pattern, C_L = 50 pF	4.5	10 8	-4 5	Module shall conform to figure 30 requirements		18,19,20, 23,24
N67	Galloping rows 1 test pattern, C_L = 50 pF	5.5	13.2	-5.5	↓	↓	↓
N68	Galloping rows 0 test pattern, C_L = 50 pF	4.5	10.8	-4 5	↓	↓	↓
N69	Galloping rows 0 test pattern, C_L = 50 pF	5.5	13.2	-5 5	↓	↓	↓
N70	Galloping columns 1 test pattern, C_L = 50 pF	4 5	10 8	-4 5	Module shall conform to figure 31 requirements		↓
N71	Galloping columns 1 test pattern, C_L = 50 pF	5.5	13.2	-5.5	↓	↓	↓
N72	Galloping columns 0 test pattern, C_L = 50 pF	4.5	10 8	-4.5	↓	↓	↓
N73	Galloping columns 0 test pattern C_L = 50 pF	5 5	13 2	-5.5	↓	↓	↓
N74	Crosstalk 1 test pattern, C_L = 50 pF	4 5	10 8	-4.5	Module shall conform to figure 32 requirements		↓
N75	↓	4.5	10 8	-5.5	↓	↓	↓
N76	↓	4.5	13 2	-4 5	↓	↓	↓
N77	↓	4 5	13.2	-5 5	↓	↓	↓
N78	↓	5.5	10.8	-4 5	↓	↓	↓
N79	↓	5 5	10.8	-5 5	↓	↓	↓
N80	↓	5 5	13 2	-4 5	↓	↓	↓
N81	↓	5.5	13.2	-5 5	↓	↓	↓
N82	Crosstalk 0 test pattern, C_L = 50 pF	4 5	10.8	-4 5	↓	↓	↓
N83	↓	4 5	10 8	-5 5	↓	↓	↓
N84	↓	4 5	13 2	-4 5	↓	↓	↓
N85	↓	4 5	13 2	-5 5	↓	↓	↓
N86	↓	5.5	10 8	-4 5	↓	↓	↓
N87	↓	5 5	10.8	-5 5	↓	↓	↓
N88	↓	5 5	13 2	-4 5	↓	↓	↓
N89	↓	5.5	13.2	-5.5	↓	↓	↓

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TABLE XIII. Dynamic functional test - Continued

Test condition	Parameter	Set item U04 voltages to ±1%			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
N90	Dynamic refresh test pattern, C_L = 50 pF	4.5	13.2	-5 5	Module shall conform to figure 33 requirements		18,19,20, 23,24
N91	Dynamic refresh test pattern C_L = 50 pF	5 5	10.8	-4.5	Module shall conform to figure 33 requirements		18,19,20, 23,24
N92	Slow read/write 1 test pattern, C_L = 50 pF	4.5	10 8	-4 5	Module shall conform to figure 34 requirements		18,19,22, 23,24
N93	Slow read/write 1 test pattern, C_L = 50 pF	5 5	13.2	-5 5	↓		↓
N94	Slow read/write 0 test pattern, C_L = 50 pF	4 5	10.8	-4.5			
N95	Slow read/write 0 test pattern, C_L = 50 pF	5.5	13 2	-5.5			
P1	Checker board 0 test pattern, C_L = 150 pF	4.5	10.8	-4.5	Module shall conform to figure 25 requirements		18,19,20, 23,24
P2	↓	4 5	10.8	-5.5	↓		↓
P3		4.5	13 2	-4.5			
P4		4.5	13.2	-5.5			
P5		5.5	10.8	-4.5			
P6		5 5	10.8	-5.5			
P7		5.5	13.2	-4 5			
P8		5.5	13.2	-5.5			
P9	Checker board 1 test pattern, C_L = 150 pF	4.5	10.8	-4.5			
P10	↓	4.5	10.8	-5.5			
P11		4.5	13 2	-4 5			
P12		4.5	13.2	-5.5			
P13		5.5	10.8	-4.5			
P14		5 5	10.8	-5.5			
P15		5.5	13.2	-4.5			
P16	↓	5 5	13 2	-5.5	↓		↓
P17	V bump test, C_L = 150 pF	See figure 26			Module shall conform to figure 26 requirements		
P18	Marching 1 test pattern, C_L = 150 pF	4 5	10.8	-4.5	Module shall conform to figure 27 requirements		18,19,21, 23,24
P19	Marching 1 test pattern, C_L = 150 pF	4 5	10 8	-5 5	Module shall conform to figure 27 requirements		18,19,21, 23,24

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TABLE XIII Dynamic functional test - Continued.

Test condition	Parameter	Set item U04 voltages to ±1%			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
P20	Marching 1 test pattern, C_L = 150 pF	4.5	13.2	-4.5	Module shall conform to figure 27 requirements		18,19,21, 23,24
P21		4.5	13.2	-5.5			
P22		5.5	10.8	-4.5			
P23		5.5	10.8	-5.5			
P24		5.5	13.2	-4.5			
P25		5.5	13.2	-5.5			
P26	Marching 0 test pattern, C_L = 150 pF	4.5	10.8	-4.5			
P27		4.5	10.8	-5.5			
P28		4.5	13.2	-4.5			
P29		4.5	13.2	-5.5			
P30		5.5	10.8	-4.5			
P31		5.5	10.8	-5.5			
P32		5.5	13.2	-4.5			
P33		5.5	13.2	-5.5			
P34	Address complement 1 test, C_L = 150 pF	4.5	10.8	-4.5	Module shall conform to figure 28 requirements		18,19,20, 23,24
P35		4.5	10.8	-5.5			
P36		4.5	13.2	-4.5			
P37		4.5	13.2	-5.5			
P38		5.5	10.8	-4.5			
P39		5.5	10.8	-5.5			
P40		5.5	13.2	-4.5			
P41		5.5	13.2	-5.5			
P42	Address complement 0 test C_L = 150 pF	4.5	10.8	-4.5			
P43		4.5	10.8	-5.5			
P44		4.5	13.2	-4.5			
P45		4.5	13.2	-5.5			
P46		5.5	10.8	-4.5			

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TABLE XIII. Dynamic functional test - Continued

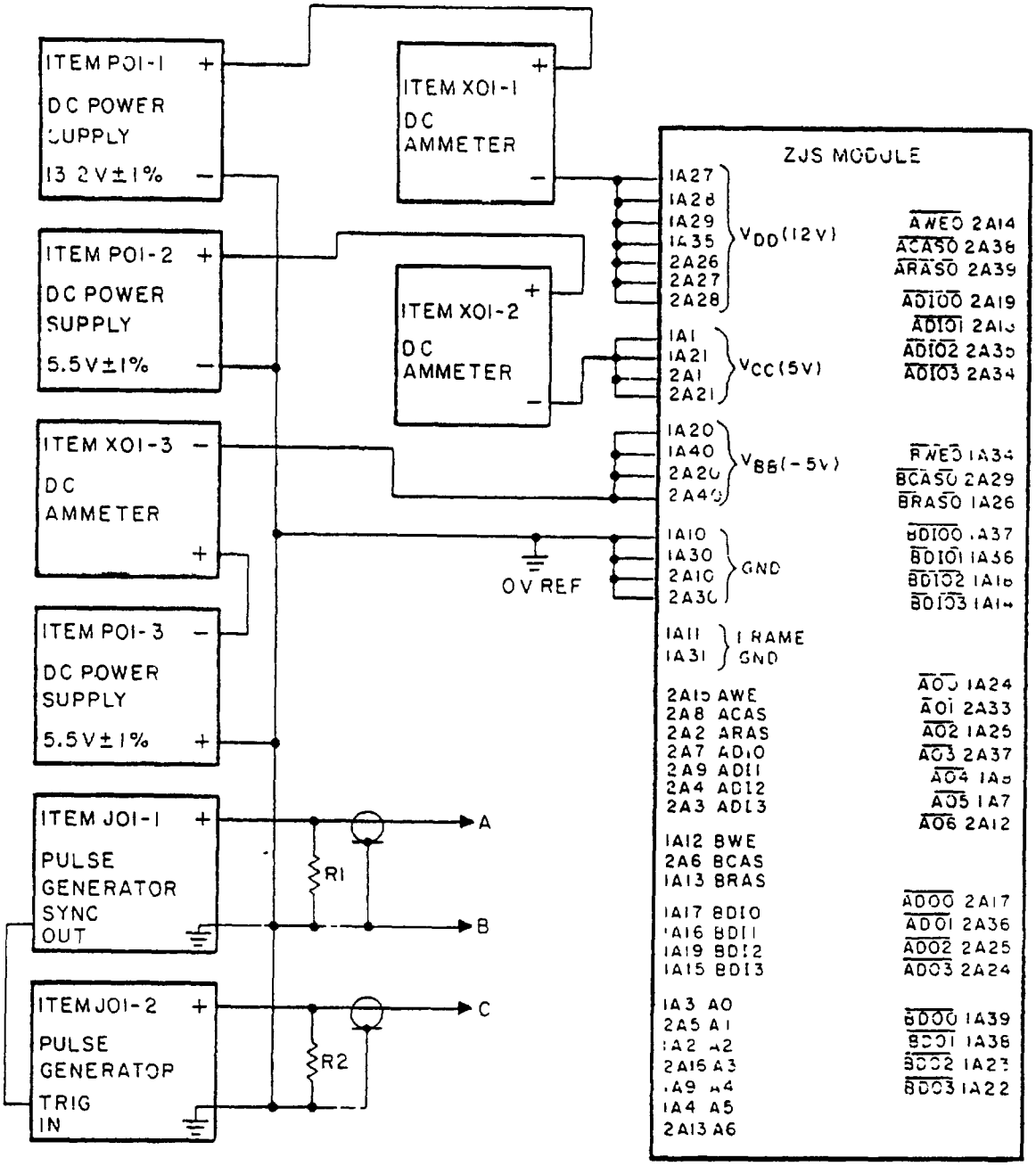
Test condition	Parameter	Set item U04 voltages to ±1%			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
P47	Address complement 0 test, C_L = 150 pF	5 5	10 8	-5 5	Module shall conform to figure 28 requirements		18,19,20, 23,24
P48	Address complement 0 test, C_L = 150 pF	5 5	13 2	-4 5	Module shall conform to figure 28 requirements		
P49	Address complement 0 test, C_L = 150 pF	5 5	13 2	-5.5	Module shall conform to figure 28 requirements		
P50	Walking diagonal 1 test pattern, C_L = 150 pF	4 5	10 8	-4 5	Module shall conform to figure 29 requirements		
P51		4.5	10.8	-5.5			
P52		4.5	13.2	-4 5			
P53		4.5	13 2	-5 5			
P54		5 5	10 8	-4 5			
P55		5 5	10 8	-5 5			
P56		5.5	13 2	-4 5			
P57		5.5	13 2	-5 5			
P58	Walking diagonal 0 test pattern C_L = 150 pF	4.5	10 8	-4.5			
P59		4 5	10 8	-5.5			
P60		4 5	13 2	-4 5			
P61		4 5	13.2	-5 5			
P62		5 5	10 8	-4 5			
P63		5 5	10 8	-5 5			
P64		5 5	13 2	-4.5			
P65		5 5	13 2	-5 5			
P66	Galloping rows 1 test pattern, C_L = 150 pF	4 5	10 8	-4 5	Module shall conform to figure 30 requirements		
P67	Galloping rows 1 test pattern, C_L = 150 pF	5 5	13 2	-5 5			
P68	Galloping rows 0 test pattern C_L = 150 pF	4 5	10 8	-4 5			
P69	Galloping rows 0 test pattern C_L = 150 pF	5.5	13 2	-5 5			
P70	Galloping columns 1 test pattern C_L = 150 pF	4 5	10 8	-4 5	Module shall conform to figure 31 requirements		

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TABLE XIII. Dynamic functional test - Continued

Test condition	Parameter	Set item U04 voltages to ±1%			Initial limits 25° ±5°C ambient		Test figure
		V_{CC} (VS2)	V_{DD} (VS1)	V_{BB} (VS3)	Min	Max	
P71	Galloping columns 1 test pattern, C_L = 150 pf	5 5	13 2	-5 5	Module shall conform to figure 31 requirements		18,19,20, 23,24
P72	Galloping columns 0 test pattern, C_L = 150 pf	4 5	10 8	-4.5	Module shall conform to figure 31 requirements		
P73	Galloping columns 0 test pattern, C_L = 150 pf	5 5	13 2	-5 5	Module shall conform to figure 31 requirements		
P74	Crosstalk 1 test pattern, C_L = 150 pF	4.5	10.8	-4 5	Module shall conform to figure 32 requirements		
P75		4 5	10.8	-5 5			
P76		4 5	13.2	-4 5			
P77		4.5	13 2	-5 5			
P78		5 5	10 8	-4 5			
P79		5.5	10 8	-5.5			
P80		5 5	13.2	-4 5			
P81		5 5	13 2	-5 5			
P82	Crosstalk 0 test pattern, C_L = 150 pF	4 5	10 8	-4 5			
P83		4 5	10 8	-5 5			
P84		4 5	13 2	-4 5			
P85		4 5	13.2	-5 5			
P86		5 5	10 8	-4 5			
P87		5 5	10 8	-5.5			
P88		5.5	13.2	-4.5			
P89		5.5	13 2	-5 5			
P90	Dynamic refresh test pattern, C_L = 150 pF	4 5	13.2	-5.5	Module shall conform to figure 33 requirements		
P91	Dynamic refresh test pattern C_L = 150 pF	5 5	10.8	-4.5	Module shall conform to figure 33 requirements		
P92	Slow read/write 1 test pattern, C_L = 150 pf	4 5	10 8	-4 5	Module shall conform to figure 34 requirements		18,19,22, 23,24
P93	Slow read/write 1 test pattern, C_L = 150 pf	5 5	13 2	-5.5			
P94	Slow read/write 0 test pattern, C_L = 150 pF	4.5	10 8	-4 5			
P95	Slow read/write 0 test pattern, C_L = 150 pf	5 5	13 2	-5.5			

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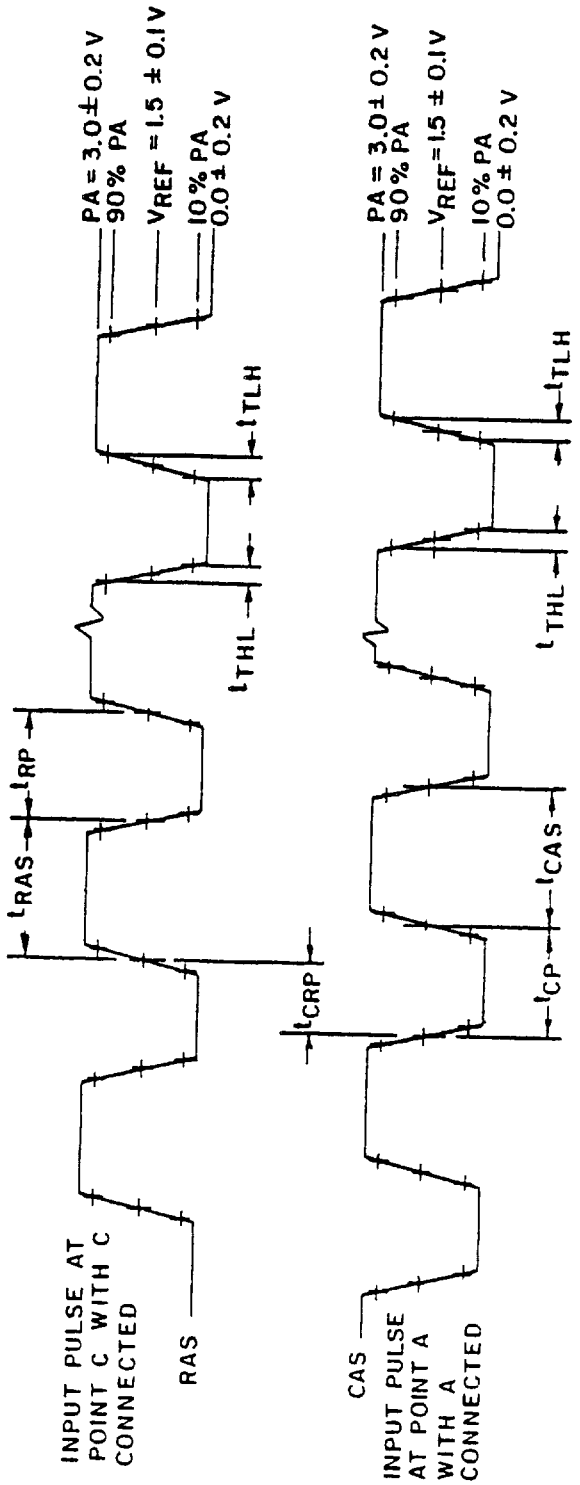
NOTES
1. R1 = R2 = 51 -5 , 1/2 W
2. Item X01-1 measures I_{DD}
3. Item X01-2 measures I_{CC}
4. Item X01-3 measures I_{BB}
5. Since ammeters X01-1, X01-2 and X01-3 are not used simultaneously, only one ammeter X01 is required

CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

FIGURE 15 Power supply current test circuit for table VII

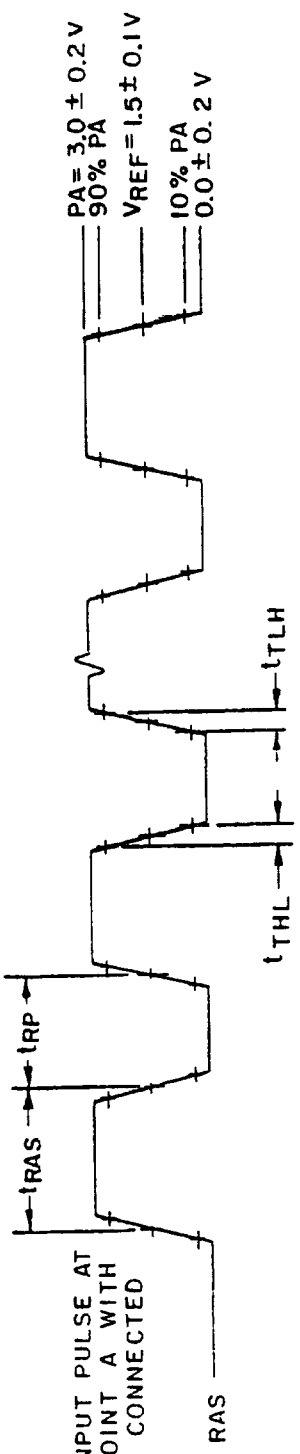
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
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NOTE Input pulse parameters
$t_{TLH} = t_{THL} = 10$ ns ±5%
t_{CRP} = 180 +5, -0 ns
t_{RAS} - 220 +5, -0 ns
t_{RP} - 180 +5, -0 ns
t_{CP} = 245 +5, -0 ns
t_{CAS} = 155 +5, -0 ns

FIGURE 16 Power supply current test circuit waveforms for table XII.

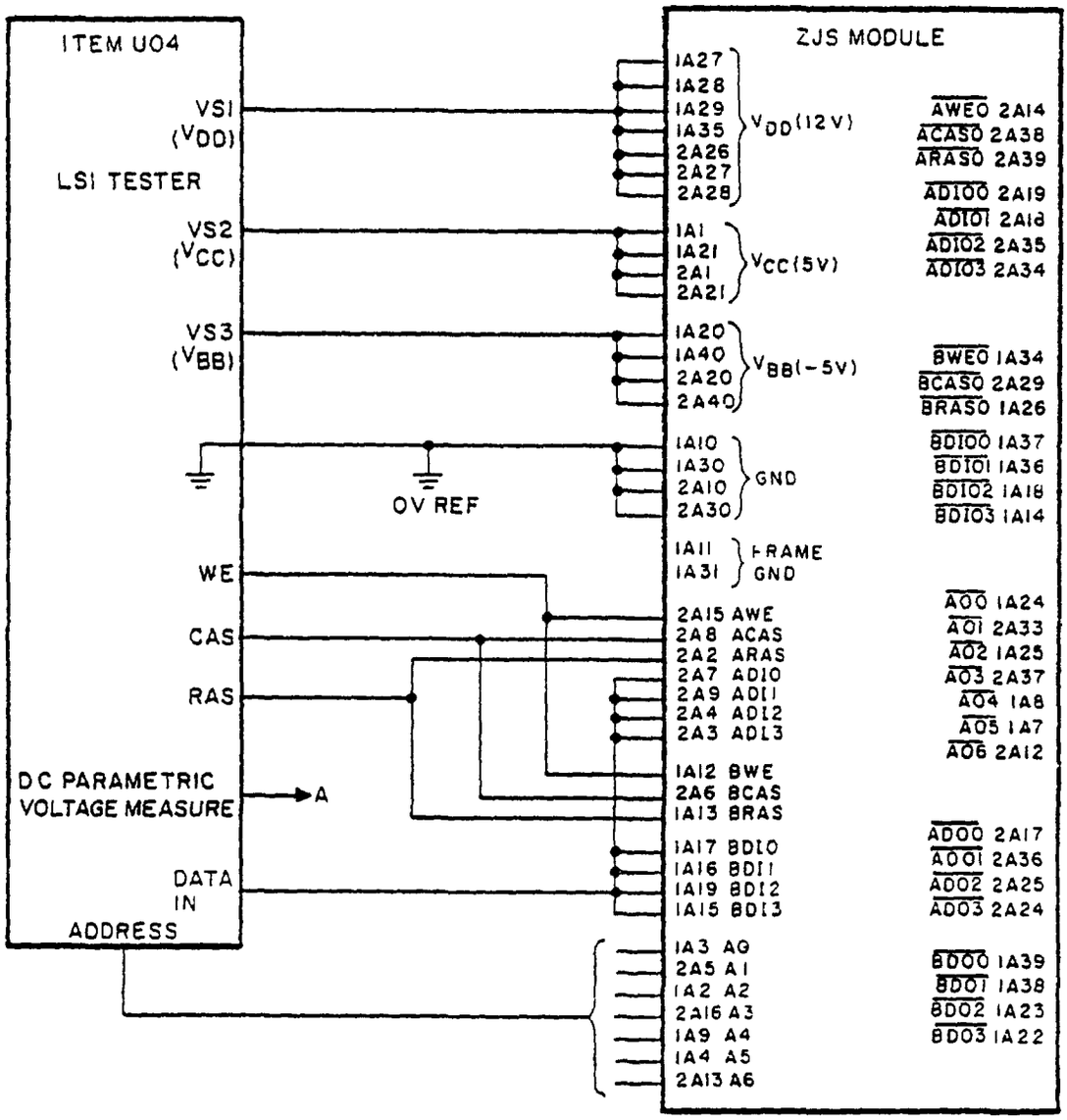
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters
$t_{TLH} = t_{THL} = 10$ ns ±5%.
$t_{RAS} = 350$ +5, -0 ns.
$t_{RP} = 150$ +5, -0 ns

FIGURE 17 Power supply current test circuit waveform for table XII.

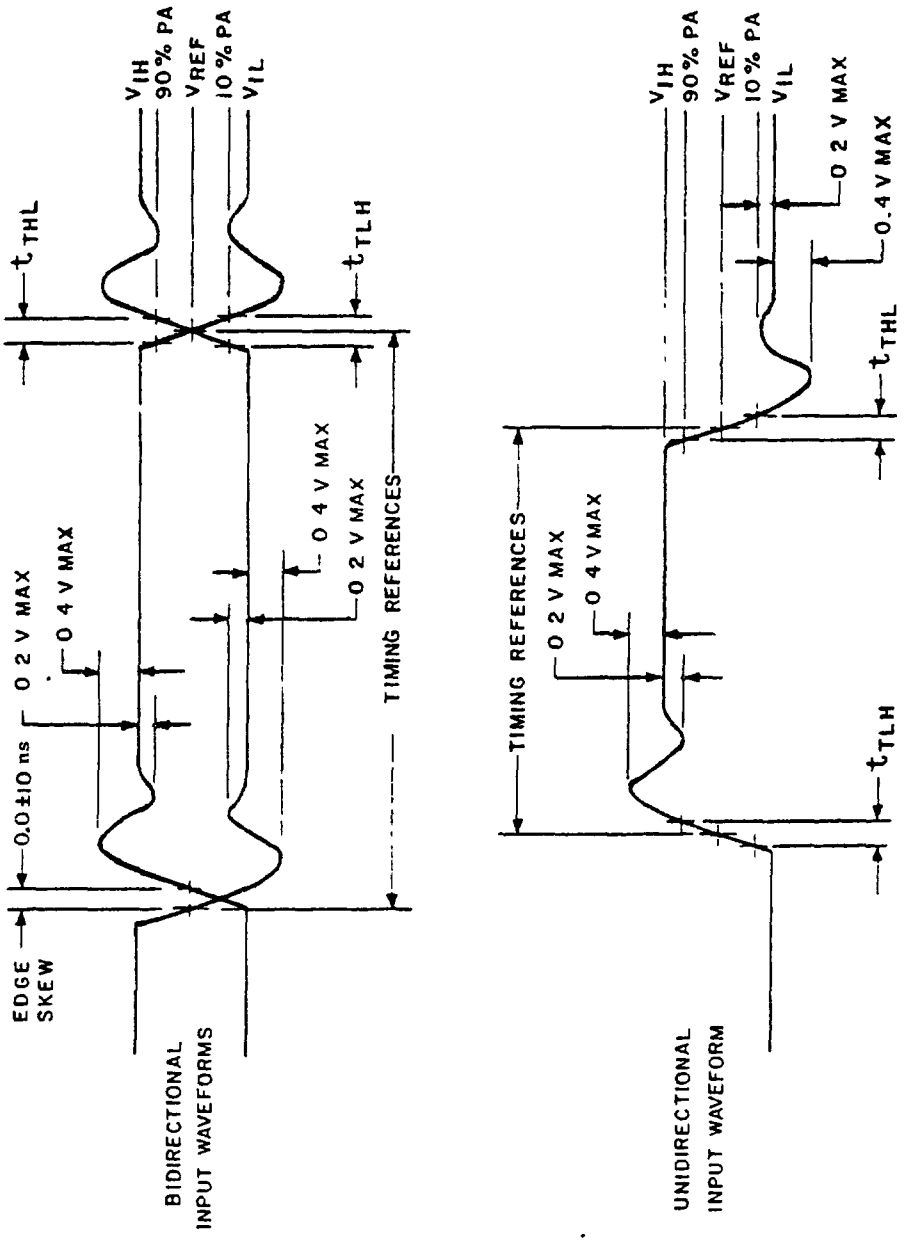
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
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NOTES
1. All tester leads to be 2 inches (50.8 mm) maximum length
2. Load each output per figure 14

FIGURE 18 Dynamic functional test circuit for table XIII

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NOTES
1 t_{TLH} - t_{THL} 10 ±2 ns
2 All timing references are made at the V_{REF} level.

FIGURE 19. Dynamic functional test circuit waveforms for table XIII.

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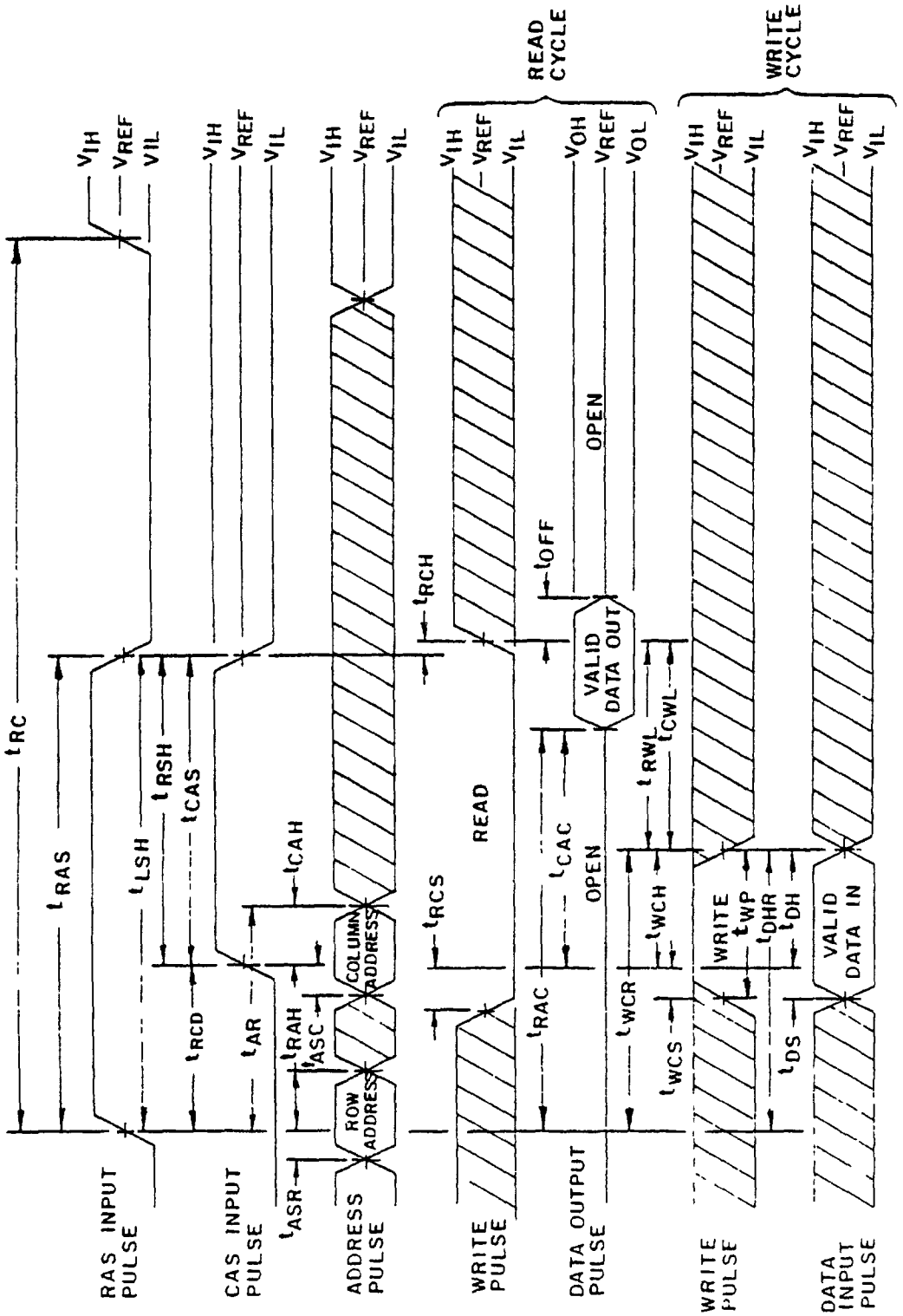


FIGURE 20 Read and write cycle timing waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
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NOTES

1. Pattern test waveform parameters
   - t_{RC} (Random read or write cycle time) = 400 +5, -0 ns.
   - t_{RAS} (RAS pulse width) = 220 +5, -0 ns.
   - t_{CAS} (CAS pulse width) = 155 +5, -0 ns.
   - t_{ASR} (Row address set-up time) = 5 +2, -0 ns.
   - t_{RAH} (Row address hold time) = 30 +2, -0 ns.
   - t_{ASC} (Column address set-up time) = 5 +2, -0 ns.
   - t_{CAH} (Column address hold time) = 60 +3, -0 ns
   - t_{RCS} (Read command set-up time) = 5 +2, -0 ns.
   - t_{WCS} (Write command set-up time) = 5 +1, -0 ns.
   - t_{WCH} (Write command hold time) = 60 +3, -0 ns.
   - t_{DS} (Data-in set-up time) = 5 +1, -0 ns.
   - t_{DH} (Data-in hold time) = 60 +3, -0 ns
   - t_{CAC} (Access time from CAS), C_L = 50 pF = 160 +5, -0 ns
   - t_{CAC} (Access time from CAS), C_L = 150 pF = 170 +5, -0 ns
   - t_{OFF} (Output buffer turn-off delay), C_L = 50 pF = 50 +0, -5 ns
   - t_{OFF} (Output buffer turn-off delay), C_L = 150 pF = 60 +0, -5 ns
   - V_{IH} = 2.0 ±0.1 V.
   - V_{IL} = 0.7 ±0.1 V.
   - V_{REF} = 1.5 ±0.1 V.
   - V_{OH} = 2.4 V minimum
   - V_{OL} = 0.4 V maximum
   - t_{RWL} = 95 +3, -0 ns.
   - t_{CWL} = 95 +3, -0 ns
   - t_{RCD} = 65 +5, -0 ns
   - t_{RAC} = 225 +5, -0 ns, (C_L = 50 pF)
   - t_{RAC} = 235 +5, -0 ns, (C_L = 150 pF)
   - t_{RSH} = 155 +5, -0 ns.
   - t_{LSH} = 220 +5, -0 ns
   - t_{AR} = 125 +5, -0 ns
   - t_{RCH} = 5 -2, -0 ns
   - t_{WCR} = 125 +5, -0 ns
   - t_{WP} = 65 +5, -0 ns
   - t_{DHR} = 125 +5, -0 ns
2. All waveforms shall conform to the detailed requirements of figure 19
3. Using the rising edge of RAS as the timing reference point, compare data out from 225 ns with C_L = 50 pF and 235 ns with C_L = 150 pF

FIGURE 20 Read and write cycle timing waveforms for table XIII - Continued

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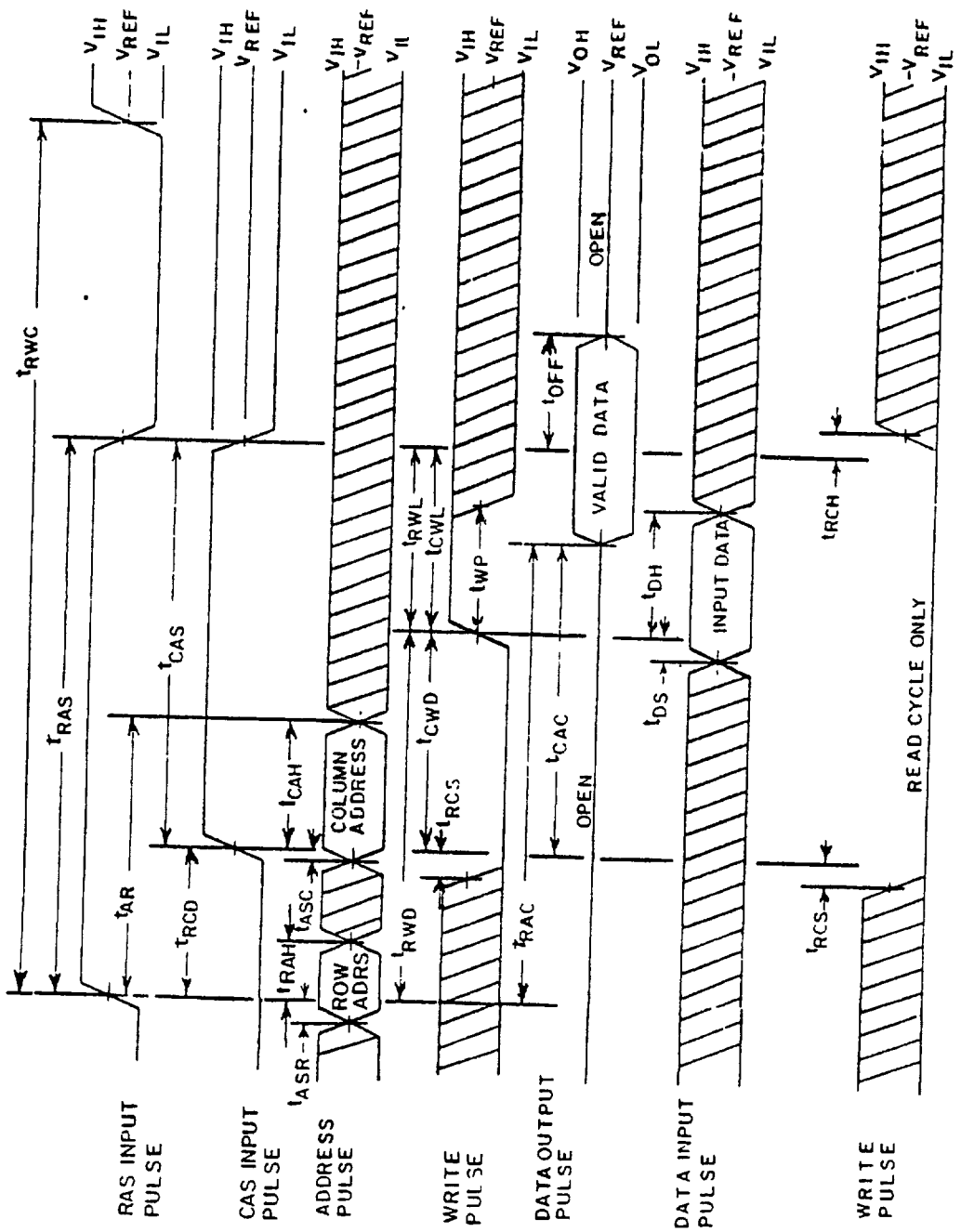


FIGURE 21. Read and write cycle timing waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES.
1. Pattern test waveform parameters:
   t_{RWC} (Random read or write cycle time) = 400 +5, -0 ns.
   t_{RAS} (RAS pulse width) = 225 +5, -0 ns.
   t_{CAS} (CAS pulse width) = 160 +5, -0 ns.
   t_{ASR} (Row address set-up time) = 5 +2, -0 ns.
   t_{RAH} (Row address hold time) = 30 +5, -0 ns.
   t_{ASC} (Column address set-up time) = 5 +2, -0 ns.
   t_{CAH} (Column address hold time) = 60 +5, -0 ns.
   t_{CWD} (CAS to write delay) = 85 +5, -0 ns.
   t_{WP} (Write command pulse width) = 65 +5, -0 ns
   t_{DS} (Data-in set-up time) = 5 +2, -0 ns.
   t_{DH} (Data-in hold time) = 65 +3, -0 ns.
   t_{RCS} (Read command set-up time) = 5 +1, -0 ns.
   t_{RCH} (Read command hold time) = 5 +1, -0 ns.
   t_{CAC} (Access time from CAS), C_L = 50 pF = 160 +5, -0 ns.
   t_{CAC} (Access time from CAS), C_L = 150 pF = 170 +5, -0 ns
   t_{OFF} (Output buffer turn-off delay), C_L = 50 pF = 50 +0, -5 ns.
   t_{OFF} (Output buffer turn-off delay), C_L = 150 pF = 60 +0, -5 ns
   V_{IH} = 2.0 ±0.1 V.
   V_{IL} = 0.7 ±0.1 V
   V_{REF} = 1.5 ±0.1 V
   V_{OH} = 2.4 V minimum.
   V_{OL} = 0.4 V maximum
   t_{AR} = 125 +5, -0 ns
   t_{RCD} = 65 +2, -0 ns.
   t_{RWL} = 75 +5, -0 ns.
   t_{CWL} = 75 +5, -0 ns
   t_{RAC} = 225 +5, -0 ns (C_L = 50 pF).
   t_{RAC} = 235 +5, -0 ns (C_L = 150 pF).
   t_{RWD} = 150 +5, -0 ns
2. All waveforms shall conform to the detailed requirements of figure 19
3. Using the rising edge of RAS as the timing reference point, compare data out from 225 ns with C_L = 50 pF and 235 ns with C_L = 150 pF

FIGURE 21. Read and write cycle timing waveforms for table VIII - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

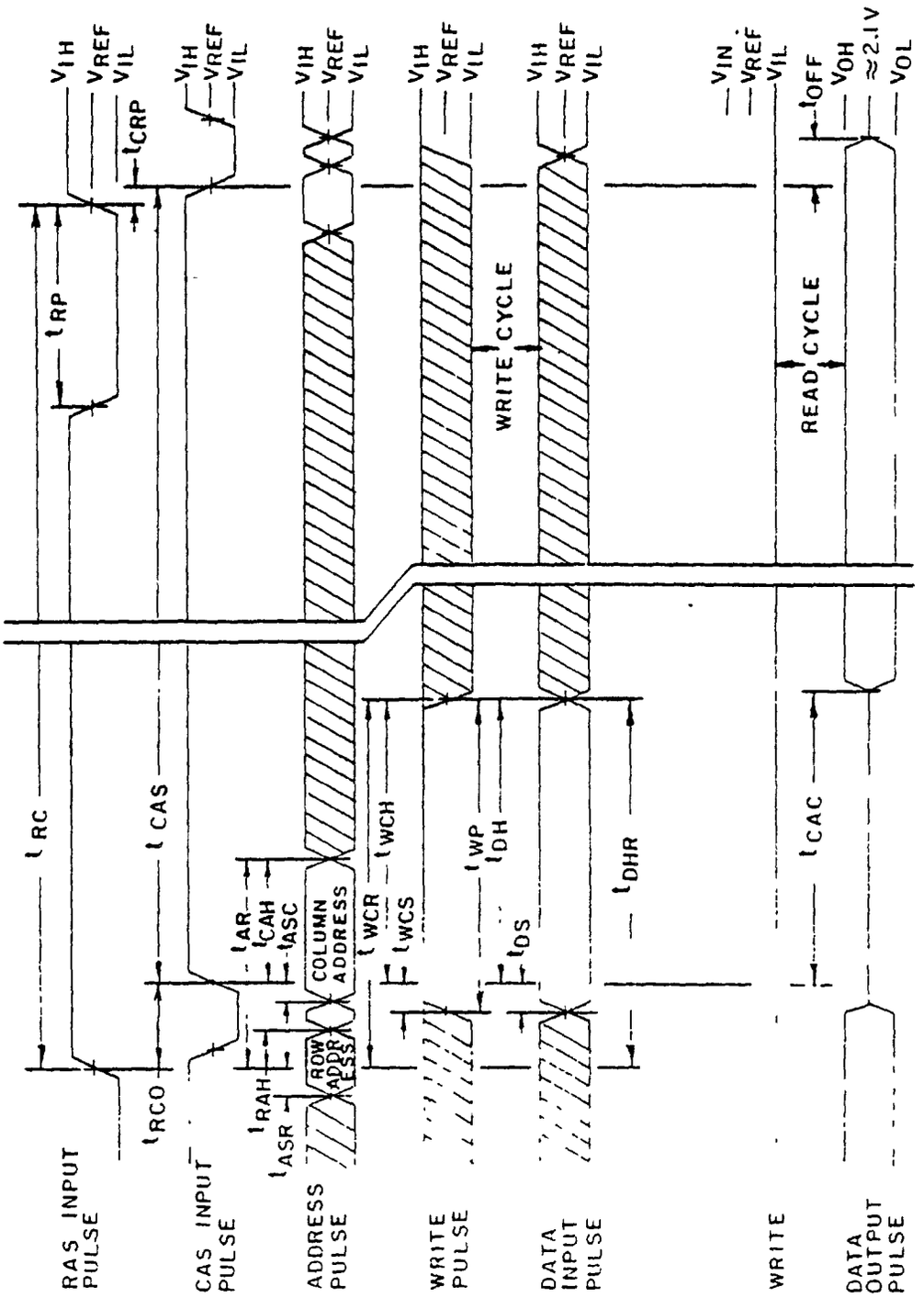


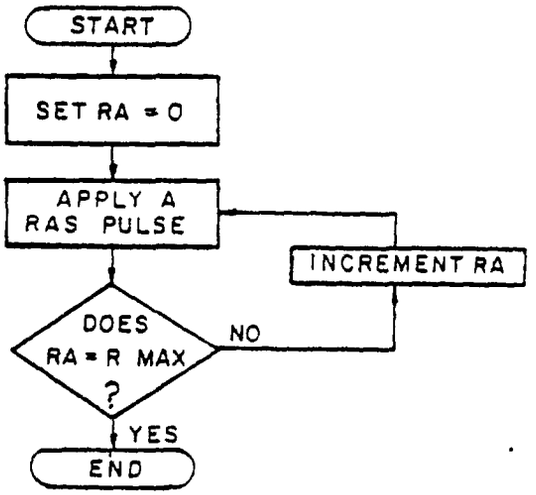
FIGURE 2. Slow read/write cycle timing waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES:
1. Pattern test waveform parameters.
   t_{RC} (Random read or write cycle time) = 5140 +5, -0 ns
   t_{RCD} (RAS to CAS delay time) = 35 +2, -0 ns.
   t_{CAS} (CAS pulse width) = 5000 +0, -5 ns.
   t_{RP} (RAS precharge time) = 140 +5, -0 ns.
   t_{ASR} (Row address set-up time) = 5 +1, -0 ns.
   t_{RAH} (Row address hold time) = 30 +2, -0 ns.
   t_{ASC} (Column address set-up time) = 5 +1, -0 ns.
   t_{WCS} (Write command set-up time) = 5 +1, -0 ns.
   t_{RCS} (Read command set-up time) = 5 +1, -0 ns
   t_{RCH} (Read command hold time) = 5 +1, -0 ns.
   t_{DS} (Data in set-up time) = 5 +1, -0 ns
   t_{CAC} (Access time from CAS), C_L = 50 pF = 5000 +5, -0 ns
   t_{CAC} (Access time from CAS), C_L = 150 pF = 5010 +5, -0 ns
   V_{IH} = 2.0 ±0.1 V.
   V_{IL} = 0.7 ±0.1 V
   V_{REF} = 1.5 ±0.1 V
   t_{AR} = 125 +5, -0 ns.
   t_{WCR} = 125 +5, -0 ns
   t_{DHR} = 125 +5, -0 ns
   t_{RAC} = 5035 +5, -0 ns
   V_{OL} = 0.4 V maximum.
   V_{OH} = 2.4 V minimum.
2. All waveforms shall conform to the detailed requirements of figure 19
3. All address lines need not switch simultaneously each cycle

FIGURE 22 Slow read/write cycle timing waveforms for table VII - Continued

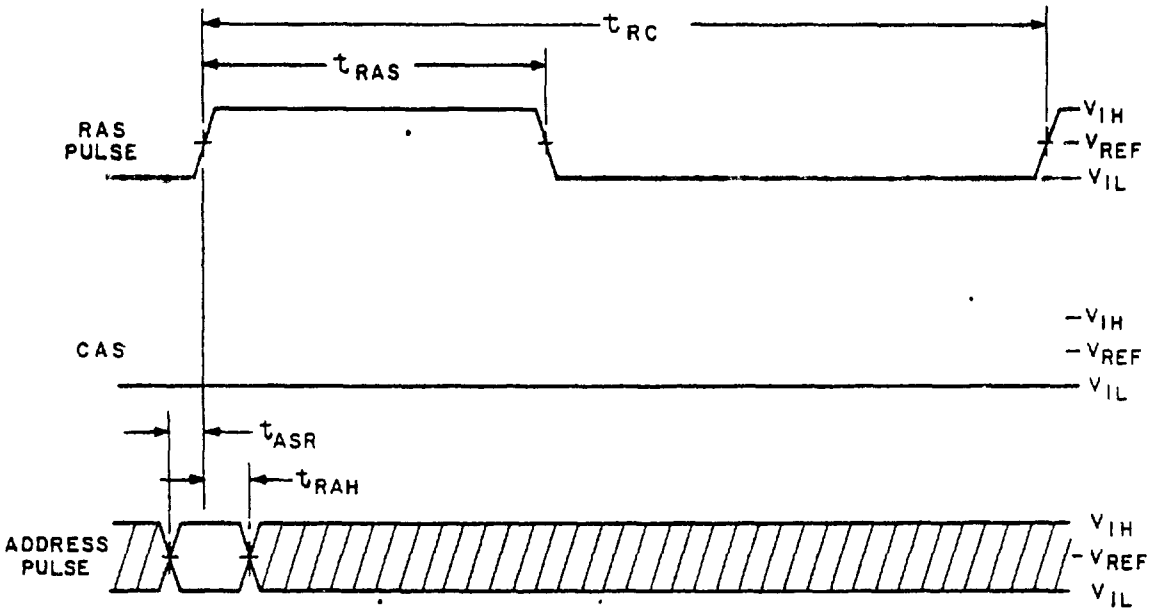
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. RA = Row address.
   RMAX = Maximum row address.
   t_{REF} = Refresh interval = 1.0 ms
2 Apply this pattern each time any row address has not been addressed in t_{REF} (1.0 ms).

FIGURE 23 Refresh flow chart for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input waveform parameters:
   t_{RC} (Random read or write cycle time) = 400 +5, -0 ns.
   t_{RAS} (RAS pulse width) = 220 +5, -0 ns.
   t_{ASR} (Row address set-up time) = 5 +1, -0 ns.
   t_{RAH} (Row address hold time) = 30 +3, -0 ns.
   V_{IH} = 2.0 ±0.1 V
   V_{IL} = 0.7 ±0.1 V
2. All waveforms shall conform to the detailed requirements of figure 19.
3. All address lines need not switch simultaneously each cycle.

FIGURE 24 Refresh flow chart waveforms for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

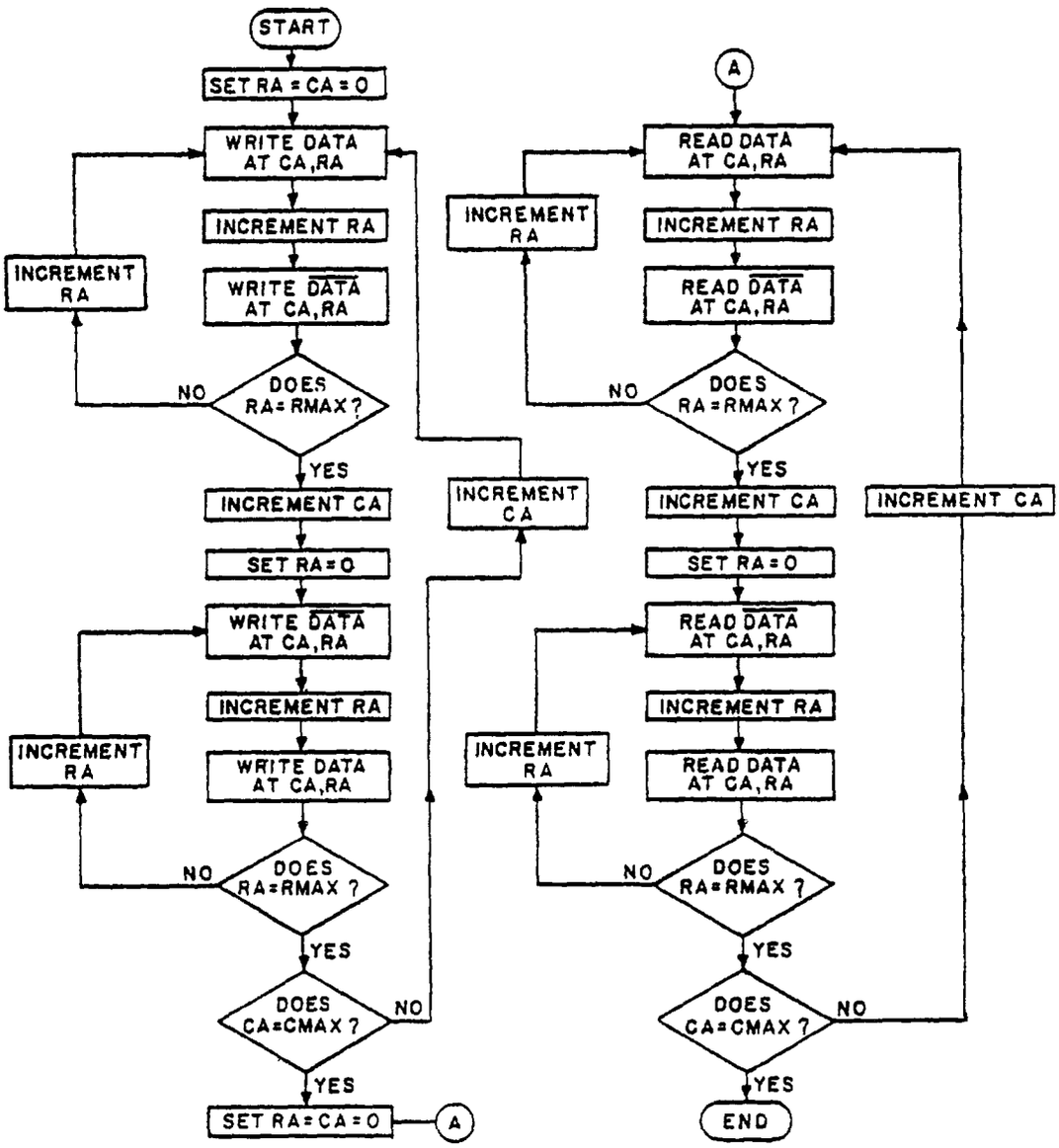


FIGURE 25 Checkerboard/checkerboard program flow chart and test program for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

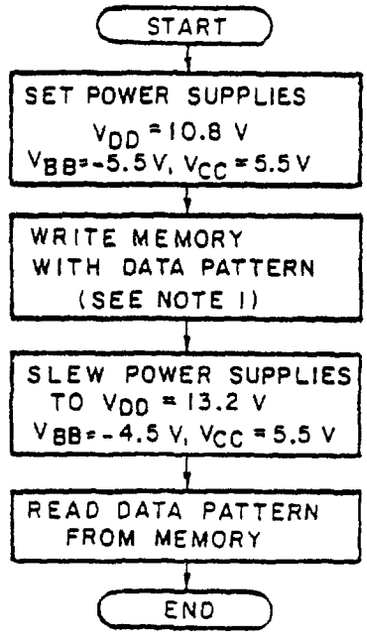
NOTES
1. RA = Row address.
   CA = Column address.
   CMAX = Maximum column address.
   RMAX = Maximum row address.
2 The objective of the checkerboard/$\overline{\text{checkerboard}}$ test pattern is to check the functionality of the module immediately after power up initialization (see note 5) and provide a quick check for "stuck at" type failures. The pattern writes a topological one-zero checkerboard data into the memory array thus providing maximum leakage paths for each cell.
3. A program that fulfills the checkerboard test pattern would

   a. Write a "high" at row 0, column 0.
   b Write a "low" at row 1, column 0.
   c. Write a "high" at row 2, column 0.
   .
   .
   d Write a "low" at row 127, column 0.
   e. Write a "low" at row 0, column 1.
   f Write a "high" at row 1, column 1.
   g Write a "low" at row 2, column 1
   .
   .
   h Write a "high" at row 127, column 1.
   i Write a "high" at row 0, column 2
   .
   .
   j Write a "high" at row 127, column 127.
   k Read a "high" at row 0, column 0.
   l Read a "low" at row 1, column 0.
   m Read a "high" at row 2, column 0
   .
   .
   .
   n. Read a "low" at row 127, column 0.
   o. Read a "low" at row 0, column 1.
   p Read a "high" at row 1, column 1
   q. Read a "high" at row 2, column 1.
   .
   .
   r Read a "high" at row 127, column 1
   s. Read a "high" at row 0, column 2.
   .
   .
   t. Read a "high" at row 127, column 127

4 A program that fulfills the $\overline{\text{checkerboard}}$ test pattern requirements would be as in note 3, with the "high" and "lows" interchanged and in the flow chart with the data and $\overline{\text{data}}$ interchanged.
5. After "power-up" the device requires an initialization sequence of eight (8) "RAS only" cycles (figure 24) to precharge the internal nodes of the dynamic circuitry to the levels required to function correctly The sequence need not be repeated during dynamic tests if power is maintained on the module continuously and the memory is not placed in standb.
6 Refresh is not required because the row address sequences through all row addresses for each increment of the column address in the flow diagram
7 The addressing sequence is based upon a topologically ideal model of a RAM Actual module tests may require address scrambling to perform the intent of these tests
8 In each "read" command in the flow chart and test program all eight outputs must be read

FIGURE 25 Checkerboard/$\overline{\text{checkerboard}}$ program flow chart and test program for table XIII - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.



NOTES.
1. The pattern required will load the memory such that all cells in the memory are in the discharge state.
2. The objective of the V_{Bump} test is to check for proper sense amplifier margins and the detection of discharged cell storage states in the present of voltage fluctuations on the power supplies. The test program discharges all memory cells in the array Then V_{DD} and V_{BB} are slewed from 10.8 to 13.2 V and -5.5 to -4.5 V respectively The memory is then read for the same pattern that was written
3. A program that fulfills the V_{Bump} test would

   a. Write each cell of the array in a pattern that satisfies note 1
   b Slew V_{DD} and V_{BB} from 10.8 to 13.2 V and from -5 5 to -4.5 V respectively
   c Read each cell of the array in a pattern that satisfies note 1

4. The power supplies are slewed in a period less than the refresh interval (t_{REF}).
5 Refresh sequencing is not required if the rows are used as the least significant bits
6 Address scrambling is not required although the data pattern must reflect any inversions in order to discharge all cells

FIGURE 26. V_{Bump} program flow chart and test program for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

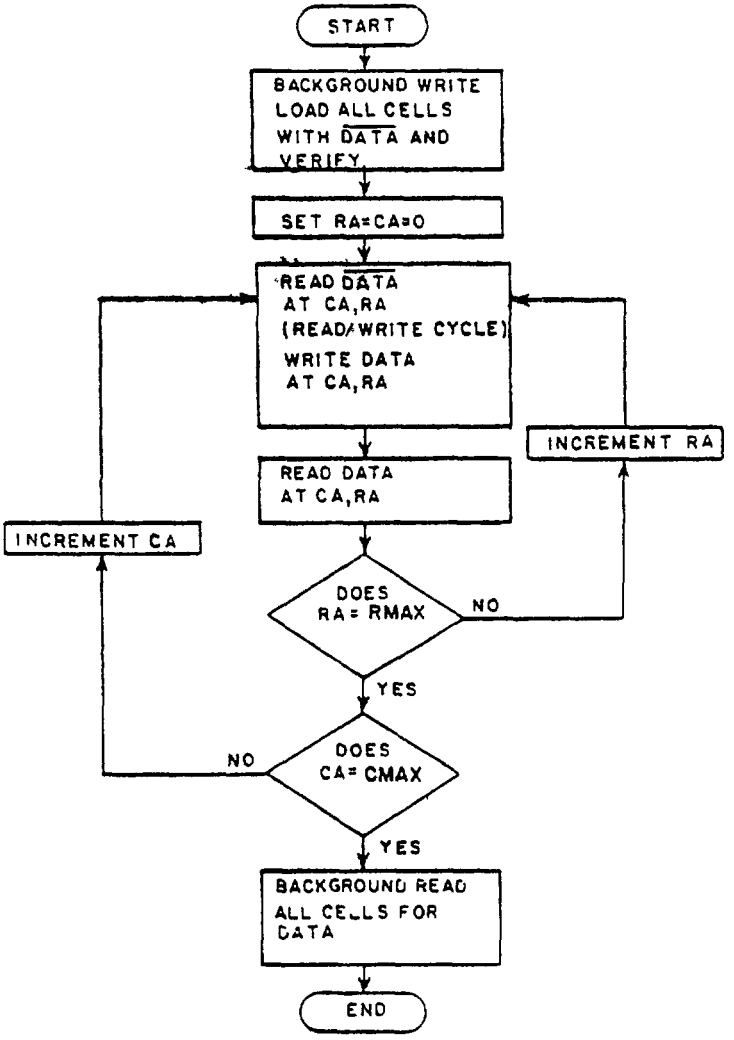


FIGURE 27. Marching-one, marching-zero program flow chart and test program for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES:
1. RA = Row address.
   CA = Column address.
   RMAX = Maximum row address.
   CMAX = Maximum column address.
2. The objective of the marching-one-marching-zero test pattern is to check for write recovery failures during a read/write cycle and provide a fast cell independence test. The pattern writes the memory to a background and performs the following sequence on each cell in the memory; Read data, write data (a read/write cycle) then read data.
3. A program that fulfills the marching-one-marching-zero test pattern would

   a. Background write to $\overline{\text{data}}$ and verify.
   b. Read a "low" at row 0, column 0.
   c. Write a "high" at row 0, column 0
   d. Read a "high" at row 0, column 0.
   e. Read a "low" at row 1, column 0.
   f. Write a "high" at row 1, column 0.

      .
      .
      .

   g. Read a "high" at row 127, column 0.
   h. Read a "low" at row 0, column 1.
   i. Write a "high" at row 0, column 1.

      .
      .

   j. Read a "high" at row 127, column 127.
   k. Background read "high" at all memory locations.

4. A program that fulfills the marching zero test pattern requirements would be as is in note 3, with the "highs" and "lows" interchanged and in note 2 and the flow chart with data and $\overline{\text{data}}$ interchanged.
5. The addressing sequence assumes a topological ideal model of a RAM. Actual module tests may require address scrambling to perform the intent of these tests.
6. If the rows are used as the least significant address bits, no refresh is required in this test pattern.

FIGURE 27. Marching-one-marching-zero program flow chart and test program for table XIII - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

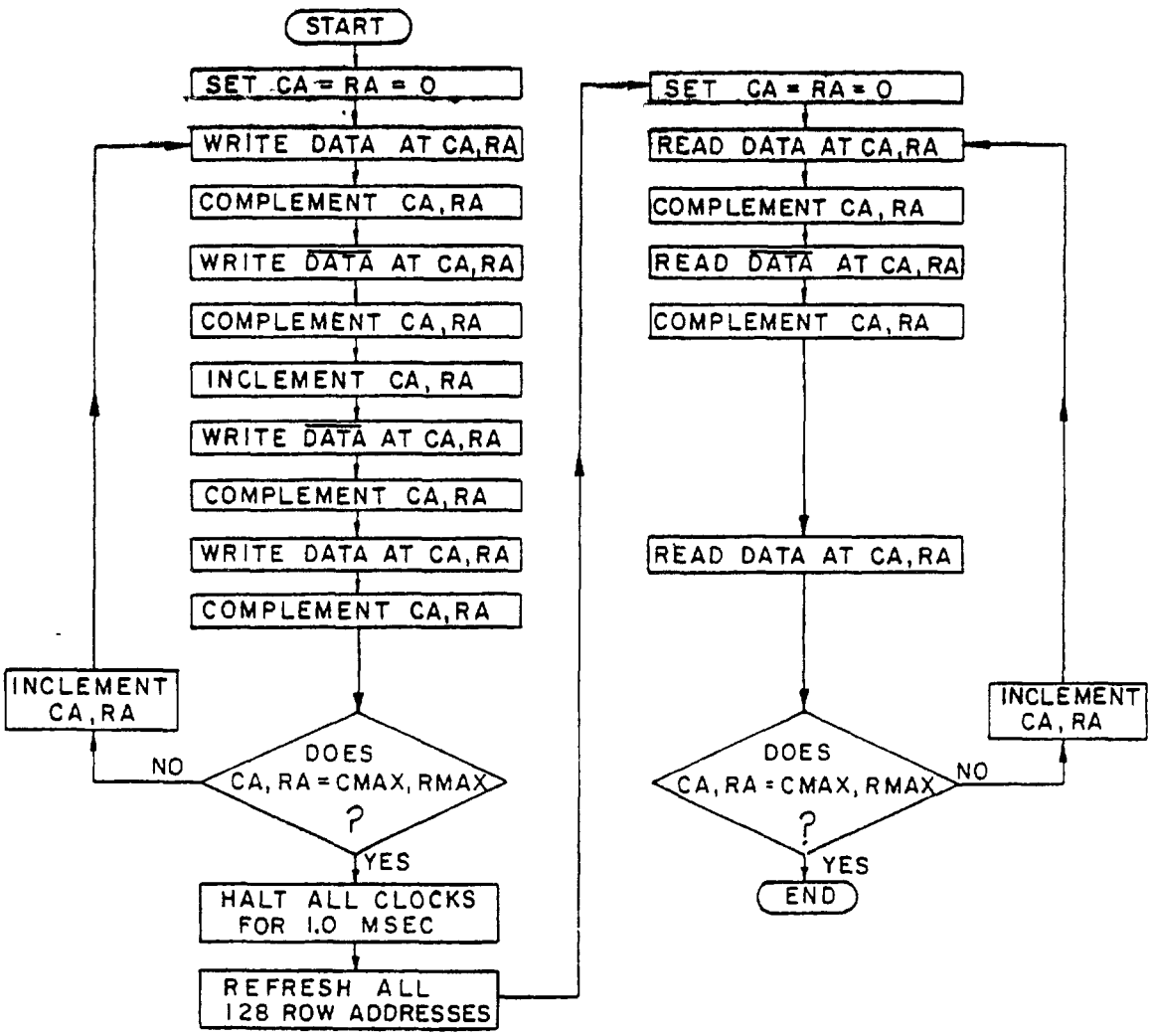


FIGURE 28. Address/Address and static refresh program flow chart and test program for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES:
1. RA = Row address.
   CA = Column address.
   RMAX = Maximum row address.
   CMAX = Maximum column address.
   CA, RA = Fourteen bit address with the row address the least significant bits.
2. The objective of the address/address and static refresh test is to generate maximum noise on the address lines. The test looks for internal address decoder problems by incorporating a test data pattern which switches the outputs each cycle. Following the load sequence all clocks are halted until the refresh interval has elapsed. The memory is then read back using the implementing address scheme.
3. A program that fulfills the address and static refresh test pattern requirements would

   a. Write a "high" at row 0, column 0.
   b. Write a "low" at row 127, column 127
   c. Write a "low" at row 1, column 0.
   d. Write a "high" at row 126, column 127.

   .

   e Write a "high" at row 0, column 1
   f Write a "low" at row 127, column 126.
   g Write a "low" at row 1, column 1

   .
   .
   .

   h Write a "high" at row 126, column 127.
   i Write a "low" at row 1, column 0.

   .
   .

   j. Read a "high" at row 0, column 0.
   k. Read a "low" at row 127, column 127.
   l. Read a "low" at row 1, column 0.
   m. Read a "high" at row 126, column 127.

   .

   .

   n Read a "high" at row 0, column 1
   o Read a "low" at row 127, column 126.
   p. Read a "low" at row 1, column 1.

   .
   .
   .

   q Read a "high" at row 126, column 127
   r Read a "low" at row 1, column 0.

4 A program that fulfills the address/address zero and static refresh test pattern requirements would be as in note 3 with the "highs" and "lows" interchanged and in the flow chart with data and data interchanged
5 The test pattern does not require address scrambling.
6 As the rows are used as the least significant bit, refresh is only required where called out in the flow chart.

FIGURE 28 Address/Address and static refresh program flow chart and test program for table XIII - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

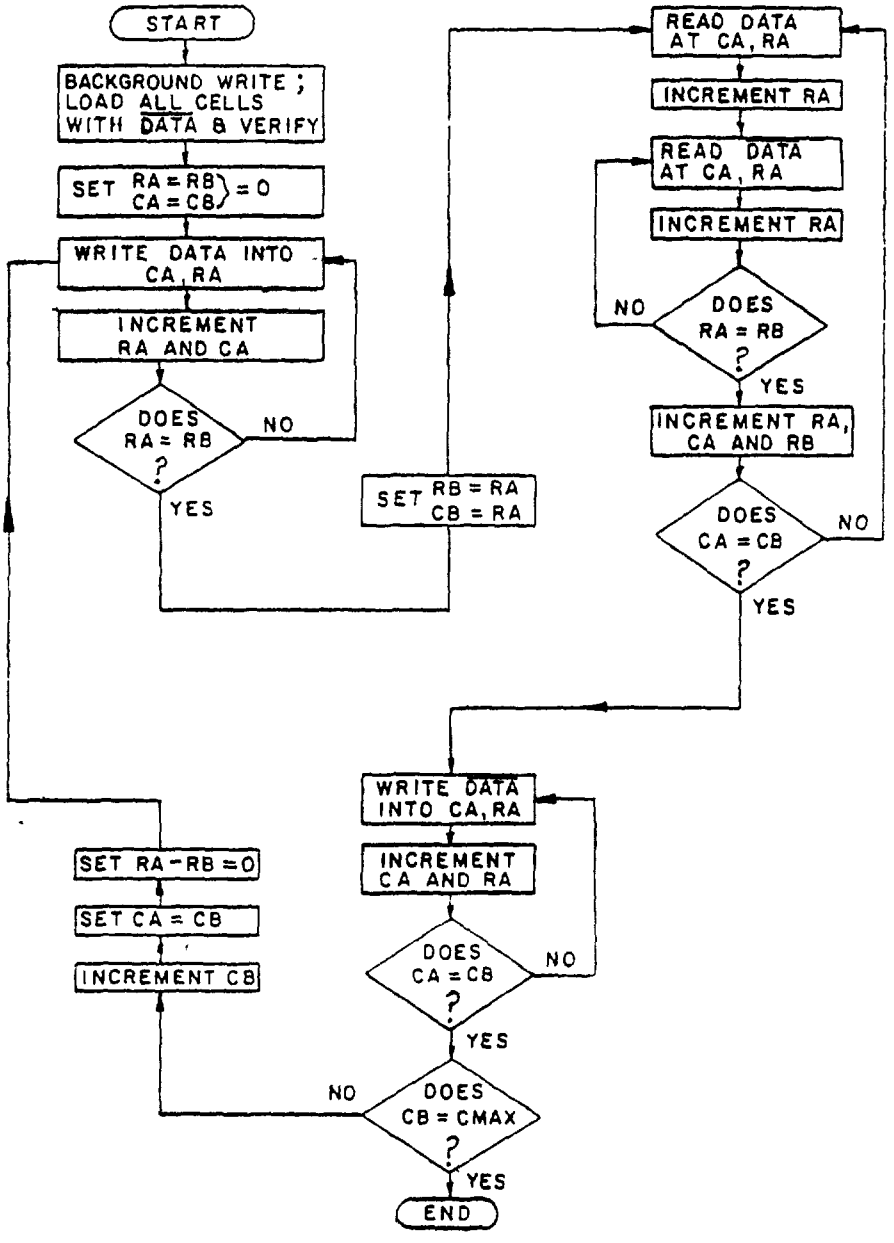


FIGURE 29 Walking diagonal-one, walking diagonal-zero program flow chart and test program for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES:
1. RA = Row address.
   CA = Column address.
   CMAX = Maximum column address.
   CB = Column base address representing the column of the first cell (row 0) in the diagonal.
   RB = Row base address representing the address of the row in each column which contains the diagonal cell.
2. The objective of the walking diagonal-one walking diagonal-zero is to establish a worse case pattern for checking for bit line inbalances, a major problem in dynamic RAMS. This pattern requires 128 write/read scans throughout the memory. On the first scan all cells are written to data with the exception of the 128 cells along the major topological diagonal which are written to data. The read scan verifies the correct operation of the array under these conditions. On each succeeding scan the position of the topological diagonal of data is shifted until on the 128th scan, it has occupied every possible position in the array. Each cell has once been the only data cell in a row and column of data
3. A program that fulfills the walking diagonal-one test pattern requirement would

   a. Background write the entire array with "lows" and verify.
   b. Write a "high" at row 0, column 0.
   c. Write a "high" at row 1, column 1.
   d. Write a "high" at row 2, column 2.

      .
      .
      .

   e. Write a "high" at row 127, column 127.
   f. Read column 0 for a "high" in row 0 and a " low" in all other cells within the column
   g. Read column 1 for a "high" in row 1 and a "low" in all other cells within the column
   h. Read column 2 for a "high" in row 2 and a "low" in all other cells within the column.

      .
      .
      .

   i. Read column 127 for a "high" in row 127 and a "low" in all other cells.
   j. Repeat sequence b. thru e. except write a "low" instead of a "high" to reset the diagonal
   k. Repeat sequence b. thru j. each time incrementing the row address by one and ending at sequence j. with the column address at 127.
   l. Write a "high" at row 0, column 127.
   m. Write a "high" at row 1, column 0.
   n. Write a "high" at row 2, column 1.

      .
      .
      .

   o. Write a "high" at row 127, column 126.
   p. Read column 127 for a "high" in row 0 and a "low" in all other cells with the column
   q. Read column 0 for a "high" in row 1, and a "low" in all other cells within the column
   r. Read column one for a "high" in row 2 and a "low" in all other cells within the column

      .
      .
      .

   s. Read column 126 for a "high" in row 127 and a "low" in all other cells within the column
   t. Repeat sequence l thru o. except write a "low" instead of the "high" to reset the diagonal.

4 Refresh is not required when the rows are used as the least significant address bits, as in the flow chart and test program.
5. The flow chart and test program are for the walking diagonal - one test To perform the walking diagonal - zero test interchange the data and data in the flow chart and the low and high in notes 2 and 3, at each reference.
6 The addressing sequence is based upon a topologically ideal model of the RAM module Tests may require address scrambling to perform the intent of these tests

FIGURE 29. Walking diagonal-one, walking diagonal-zero program flow chart and test program for table XIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

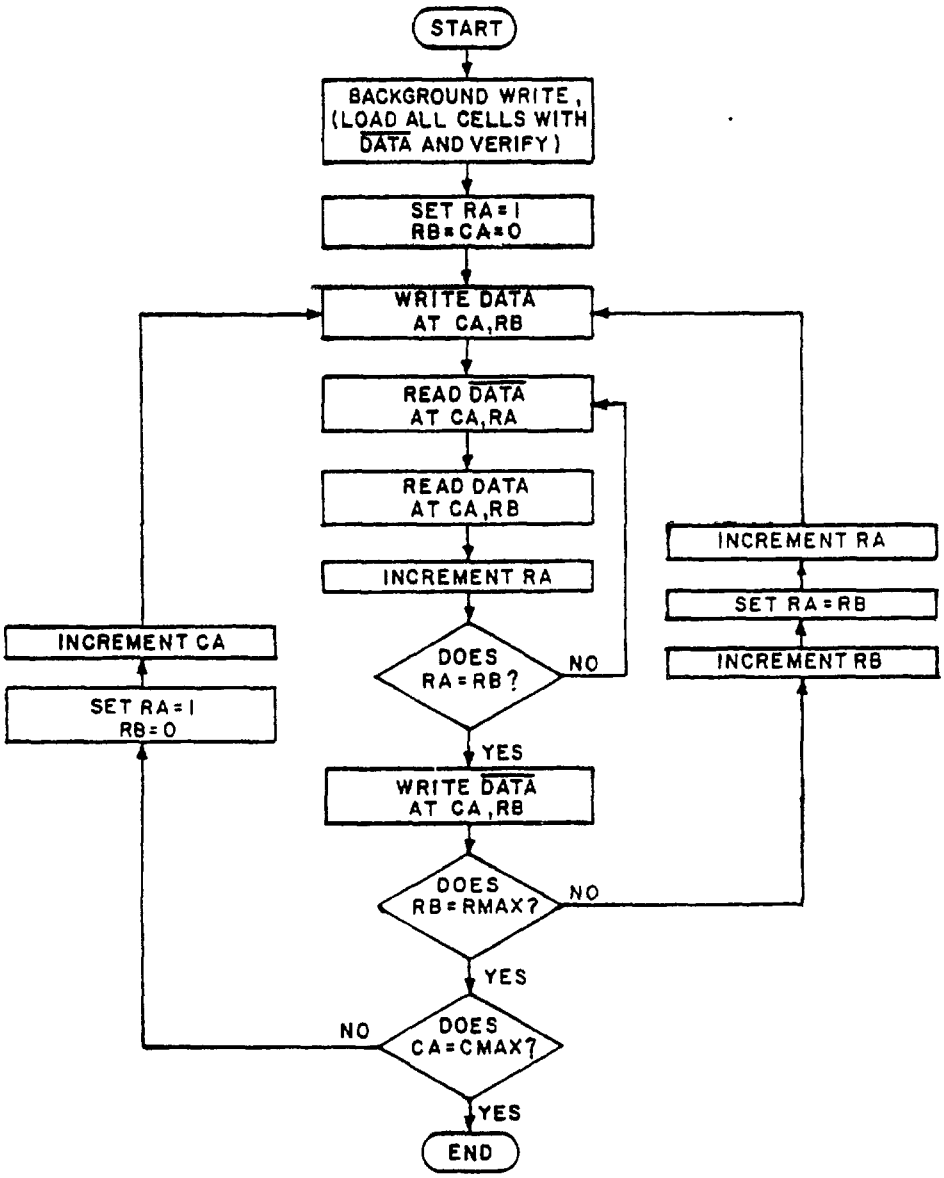


FIGURE 30 Galloping rows-one, galloping rows-zero program flow chart and test program for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES:
1. RA = Row address.
   CA = Column address.
   RB = Row base address representing the row address of the galloping data bit.
   RMAX = Maximum row address.
   CMAX = Maximum column address.
2. The objective of the galloping rows-one-galloping rows-zero test pattern is to check for row line to column line crosstalk, which can upset the data flow within the memory. In this test a standard galloping address pattern is executed on each cell of the array, with respect to all other cells in the same topological column.
3. A program that fulfills the galloping rows-one test pattern would:

   a. Background write "lows" in all memory cells and verify.
   b. Write a "high" at row 0, column 0.
   c. Read a "low" at row 1, column 0.
   d. Read a "high" at row 0, column 0.
   e. Read a "low" at row 2, column 0.
   .
   .
   .
   f. Read a "low" at row 127, column 0.
   g. Write a "low" at row 0, column 0.
   h. Write a "high" at row 1, column 0.
   i. Read a "low" at row 2, column 0
   j. Read a "high" at row 1, column 0.
   .
   .
   .
   k. Read a "low" at row 127, column 0.
   l. Read a "low" at row 0, column 0.
   .
   .
   .
   m. Write a "high" at row 127, column 0.
   .
   n. Read a "high" at row 127, column 0
   o. Repeat sequences b thru n. in column 1
   .
   .
   .
   p. Repeat sequences b thru n. in column 127

4 A program that fulfills the galloping rows-zero test pattern requirements would be as in note 3 with the "highs" and "lows" interchanged and the flow chart with the data and data interchanged
5 The addressing sequence assumes a topological ideal model of the RAM. Actual module tests may require address scrambling to perform the intent of these tests.
6. As the rows are used as the least significant bits, no refresh is required in this test pattern

FIGURE 30 Galloping rows-one, galloping rows-zero program flow chart and test program for table XIII - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

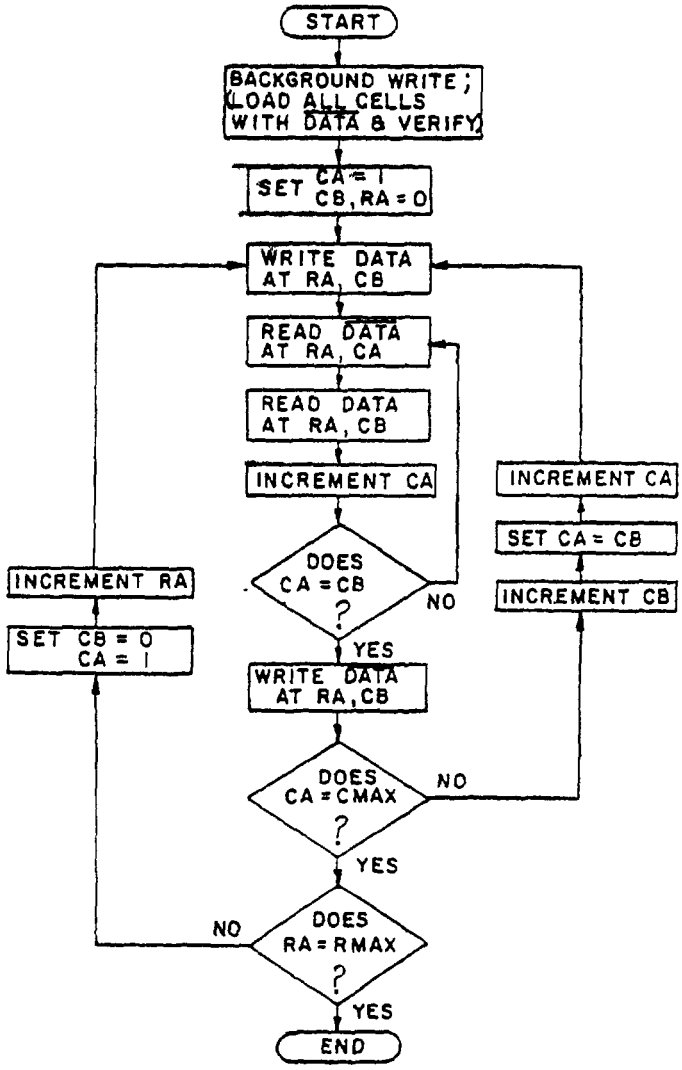


FIGURE 31 Galloping columns-one, galloping columns-zero program flow chart and test program for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES.
1. RA = Row address.
   CA = Column address.
   CB = Column base address representing the column address of the galloping data bit.
   CMAX=Maximum column address.
   RMAX = Maximum row address.
2. The objective of the galloping columns-one-galloping columns-zero test pattern is to check for column line to row line crosstalk, which can cause multiple bit selection within the memory array. In this test a standard galloping pattern is executed on each cell of the array with respect to all other cells in the same topological row.
3. A program that fulfills the galloping columns-one test pattern would:

   a. Background write "low" in all memory cells and verify.
   b. Write a "high" at row 0, column 0.
   c. Read a "low" at row 0, column 1.
   d Read a "high" at row 0, column 0.
   e. Read a "low" at row 0, column 2.
   .
   .
   .
   f Read a "low" at row 0, column 127.
   g. Write a "low" at row 0, column 0.
   h. Write a "high" at row 0, column 1
   i. Read a "low" at row 0, column 2.
   j. Read a "high" at row 0, column 1.
   .
   .
   .
   k Read a "low" at row 0, column 127.
   l. Read a "low" at row 0, column 0.
   .
   .
   m. Write a "high" at row 0, column 127.
   .
   n. Write a "low" at row 0, column 0.
   o. Repeat sequences b. thru n. in row 1.
   .
   .
   p. Repeat sequences b thru n. in row 127

4. A program that fulfills the galloping columns-zero test pattern requirements would be as in note 3 with the "highs" and "lows" interchanged and in the flow chart with the data and data interchanged.
5. The addressing sequence assumes a topological ideal model of the RAM Actual module tests may require address scrambling to perform the intent of these tests
6. The module must be refreshed every refresh interval as the columns are used as the least significant address bits.

FIGURE 31. Galloping columns-one, galloping columns-zero program flow chart and test program for table XIII - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

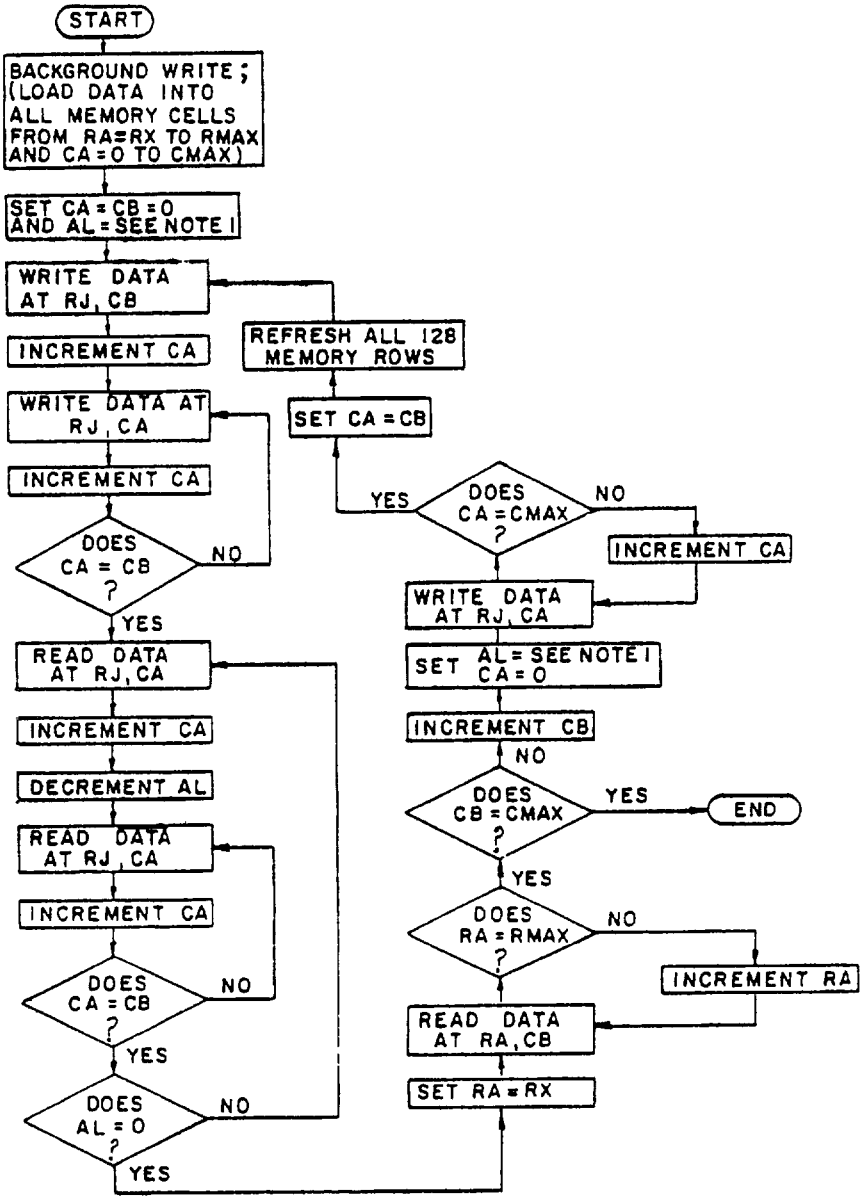


FIGURE 32 Crosstalk-one, crosstalk-zero program flow chart and test program for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:32Z
Check the source to verify that this is the current version before use.

NOTES

1. RA = Row address.
   CB = Column base address.
   RJ = Row J (see note 2).
   RX, RMAX = Beginning and ending row address in 1/2 the memory

$$AL = \frac{(t_{REF})}{(t_{RC}) \times 128 \text{ cells/row}} - 1.0$$

   Truncate AL to a whole (nondecimal) number to avoid reading row J beyond the refresh minimum time.
2. Pass 1, RX = Zero, RMAX = 63, RJ = 64.
   Pass 2, RX = 64, RMAX = 127, RJ = Zero.
   The flow chart must be run with both pass setups to complete the crosstalk-one test To perform the crosstalk-zero test interchange the "high" and "low" for each reference in the test program and data and data for each reference in the flow chart and complete both passes
3. The objective of the crosstalk-one-crosstalk-zero test is to produce worse case noise conditions, across column lines of the memory, which can degrade stored charge. The test program sets up a pattern in memory such that in one half of the memory a check bit is written to the same polarity as other half of the memory and all remaining cells in the check bit row are written to the opposite polarity The check bit row is repeatedly read until the refresh period (t_{REF}) has elapsed. Whereupon the column containing the check bit in the other half of the memory is read to verify stored data This sequence is repeated until every cell in the check bit row has once been the check bit.
4. A program that fulfills the crosstalk-one, pass 1 test pattern requirements would·

   a. Write a "high" in all cells in rows 0 to 63
   b Write a "high" at row 64 column 0.
   c. Write a "low" at row 64 and columns 1 to 127.
   d. Read row 64 until the refresh period (t_{REF}) has elapsed.
   e. Read a "high" in all cells of column zero, row zero to 63.
   f. Write a "high" at row 64, column one.
   g. Write a "low" at row 64, column 0, 2, 3, to 127.
   h. Read row 64 until the refresh period (t_{REF}) has elapsed.
   i Read a "high" in all cells of column 1, row 0 to 63

   .
   .
   .

   j. Write a "high" at row 64 column 127
   k. Write a "low" at row 64 columns 0 to 126
   l Read row 64 until the refresh period (t_{REF}) has elapsed
   m. Read a "high" in all cells of column 127, rows 0 to 63.
   n. end.

5. The address sequence assumes a topologically ideal model of the RAM. Actual module tests may require address scrambling to perform the intent of these tests

FIGURE 32 Crosstalk-one, crosstalk-zero program flow chart and test program for table XIII - Continued.

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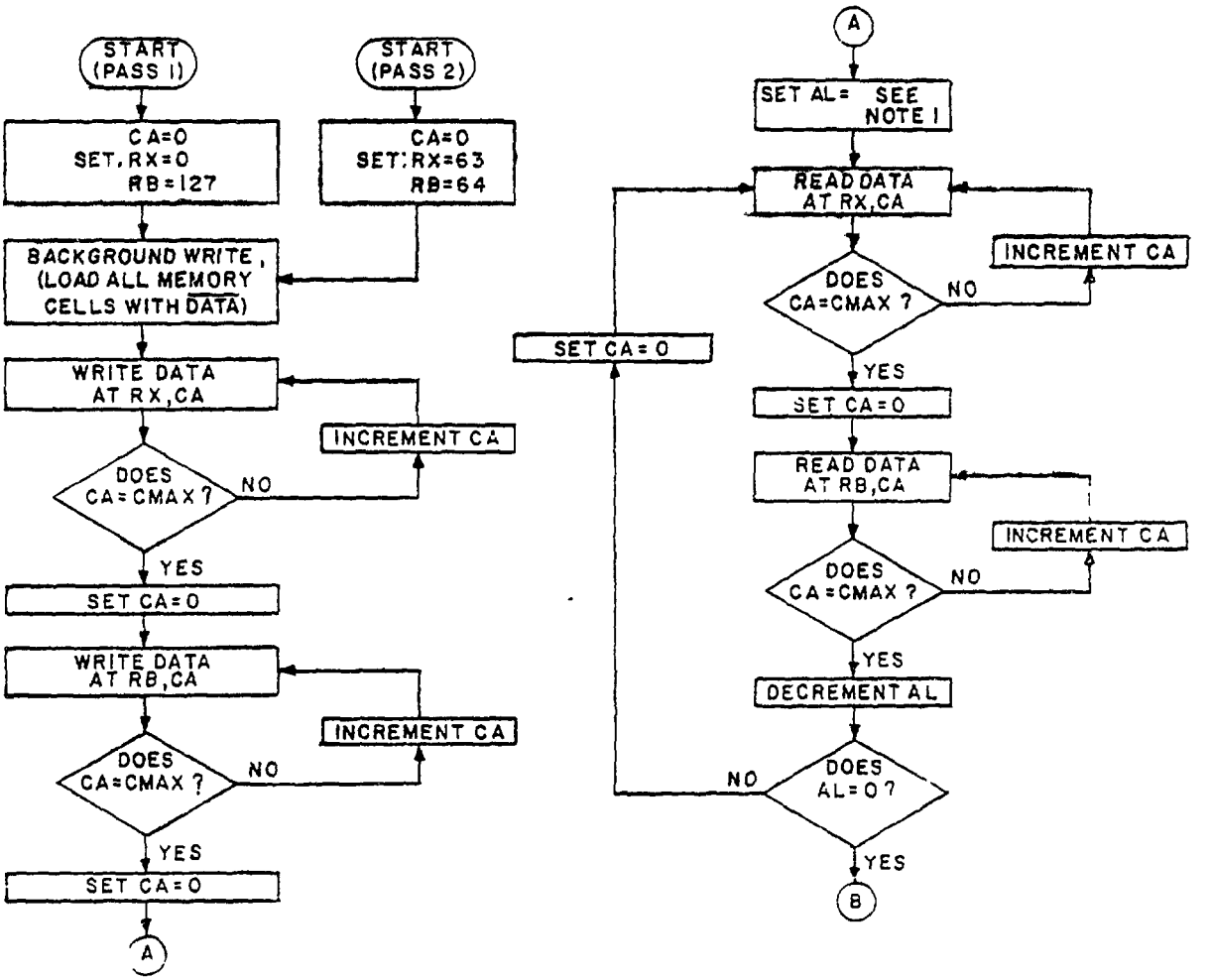


FIGURE 33 Dynamic refresh program flow chart and test program for table XII.

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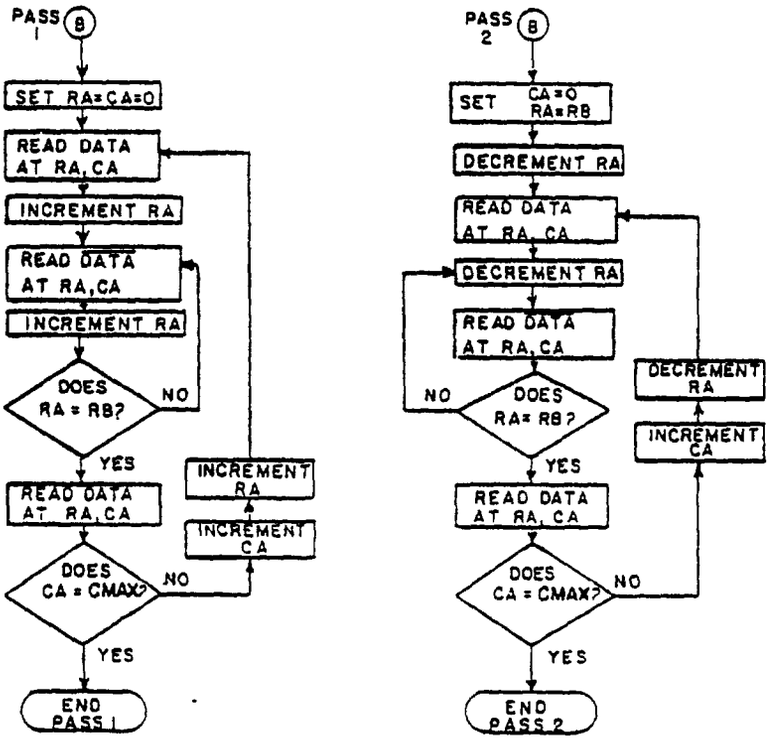


FIGURE 33. Dynamic refresh program flow chart and test program for table XIII - Continued

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NOTES.

1. CA = Column address.
   CMAX = Maximum column address.
   RA = Row address.
   RMAX = Maximum row address.
   RX, RB are 4 rows in the memory specifically used in this test, (see note 2).

$$AL = \left[\frac{(t_{REF})}{(t_{RC}) \times 128 \text{ cells/row} \times 2 \text{ rows}} \right] - 1.0$$

   Truncate AL to a whole number

2. The object of the dynamic refresh test is an attempt to degrade stored zeros by continuously reading of two rows of "ones", one row in each half of the memory. The memory is first written to all "zero", and then two test rows (topological rows 0 and 127) are written to ones. The test rows are read alternately until the refresh period has elapsed, and then the entire array is read. Two new test rows are chosen (topological rows 63 and 64) and the same process is repeated. The continuous reading of "ones" can degrade the stored zeros from restore noise problems.
3. A program that fulfills the dynamic refresh test pattern requirements would:

   a. Write "lows" in all memory locations (Pass 1).
   b. Write "highs" in row 0.
   c. Write "highs" in row 127
   d. Read row 0 then row 127 then row 0 etc. until the refresh period has elapsed.
   e. Read the whole memory for "highs" in rows 0 and 127 and "lows" in all other rows
   f. Begin Pass 2.
   g. Write "lows" in all memory locations.
   h. Write "highs" in row 63
   i. Write "highs" in row 64.
   j. Read "highs" in row 63, then row 64, then row 63 until the refresh period has elapsed.
   k. Read the memory for "highs" in rows 63 and 64 and "lows" in all other rows.
   l. End.

4. Refresh is required prior to steps d. and j in the test program in order to set up maximum time for reading the two test rows.
5. The address sequence is based upon a topologically ideal model of a RAM. Actual module tests may require address scrambling to perform the intent of the tests

FIGURE 33 Dynamic refresh program flow chart and test program for table XIII - Continued

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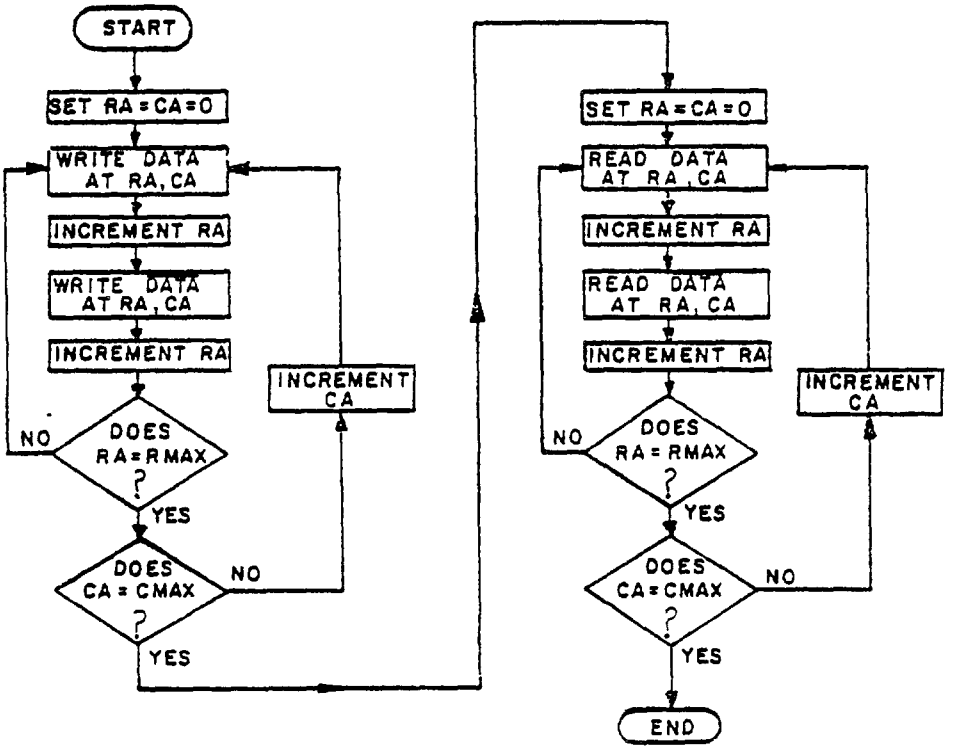


FIGURE 34. Slow read/write-one, slow read/write-zero program flow chart and test program for table XIII

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NOTES:
1. RA = Row address.
   RMAX = Maximum row address.
   CA = Column address.
   CMAX = Maximum column address.
2. The objective of the slow read/write-one, slow read/write-zero test pattern is to check the functionality of the module when used with maximum RAS and CAS pulse widths. The program writes each column with data then proceeds to the next column. After the memory is written, the entire memory is read using the same RAS/CAS timing.
3. A program that fulfills the slow read/write-one test pattern requirements would:

   a. Write a "high" at row 0, column 0.
   b. Write a "low" at row 1, column 0.
   c. Write a "high" at row 2, column 0

   .
   .
   .

   d Write a "low" at row 127, column 0.
   e Write a "high" at row 0, column 1.
   f. Write a "low" at row 1, column 1.
   g. Write a "high" at row 2, column 1.

   .
   .
   .

   h. Write a "low" at row 127, column 127.
   i. Read a "high" at row 0, column 0.
   j. Read a "low" at row 1, column 0.

   .
   .
   .

   k Read a "low" at row 127, column 0.
   l Read a "high" at row 0, column 1
   m. Read a "low" at row 1 column 1.

   .
   .
   .

   n. Read "low" at row 127, column 127.

4 A program that fulfills the slow read/write-zero test pattern would be as in note 3 with the "highs" replaced by "lows" and in the flow chart with the data replaced by data
5. Address scrambling is not required for the write and read address sequence.

FIGURE 34 Slow read/write-one, slow read/write-zero program flow chart and test program for table XIII - Continued.

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test.

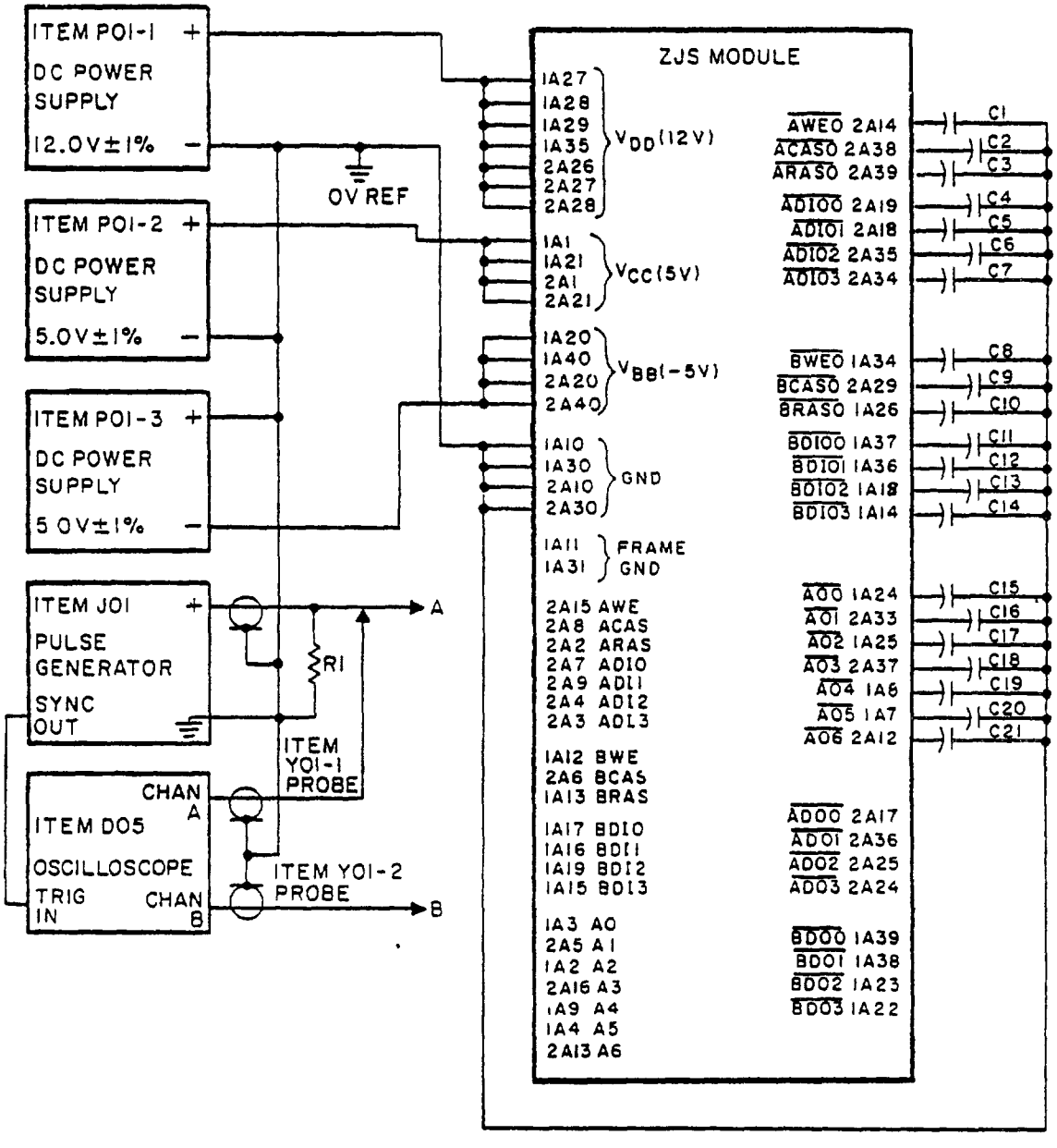
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Q1	$\overline{AWE0}$	Propagation delay time, (t_{PLH}), C_L = 50 pF	2A15	2A14	3	30	ns	35,36
Q2	$\overline{ACAS0}$		2A8	2A38				
Q3	$\overline{ARAS0}$		2A2	2A39				
Q4	$\overline{ADIO0}$		2A7	2A19				
Q5	$\overline{ADIO1}$		2A9	2A18				
Q6	$\overline{ADIO2}$		2A4	2A35				
Q7	$\overline{ADIO3}$		2A3	2A34				
Q8	$\overline{BWE0}$		1A12	1A34				
Q9	$\overline{BCAS0}$		2A6	2A29				
Q10	$\overline{BRAS0}$		1A13	1A26				
Q11	$\overline{BDIO0}$		1A17	1A37				
Q12	$\overline{BDIO1}$		1A16	1A36				
Q13	$\overline{BDIO2}$		1A19	1A18				
Q14	$\overline{BDIO3}$		1A15	1A14				
Q15	$\overline{AO1}$		1A3	1A24				
Q16	$\overline{AO2}$		2A5	2A33				
Q17	$\overline{AO3}$		1A2	1A25				
Q18	$\overline{AO4}$		2A16	2A37				
Q19	$\overline{AO5}$		1A9	1A8				
Q20	$\overline{AO6}$		1A4	1A7				
Q21	$\overline{AO7}$		2A13	2A12				

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TABLE XIV Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	ā to pin	Min	Max		
R1	$\overline{AWE0}$	Propagation delay time, (t_{PHL}), C_L = 50 pF	2A15	2A14	3	30	ns	35,36
R2	$\overline{ACAS0}$		2A8	2A38				
R3	$\overline{ARAS0}$		2A2	2A39				
R4	$\overline{ADIO0}$		2A7	2A19				
R5	$\overline{ADIO1}$		2A9	2A18				
R6	$\overline{ADIO2}$		2A4	2A35				
R7	$\overline{ADIO3}$		2A3	2A34				
R8	$\overline{BWE0}$		1A12	1A34				
R9	$\overline{BCAS0}$		2A6	2A29				
R10	$\overline{BRAS0}$		1A13	1A26				
R11	$\overline{BDIO0}$		1A17	1A37				
R12	$\overline{BDIO1}$		1A16	1A36				
R13	$\overline{BDIO2}$		1A19	1A18				
R14	$\overline{BDIO3}$		1A15	1A14				
R15	$\overline{AO1}$		1A3	1A24				
R16	$\overline{AO2}$		2A5	2A33				
R17	$\overline{AO3}$		1A2	1A25				
R18	$\overline{AO4}$		2A16	2A37				
R19	$\overline{AO5}$		1A9	1A8				
R20	$\overline{AO6}$		1A4	1A7				
R21	$\overline{AO7}$		2A13	2A12				

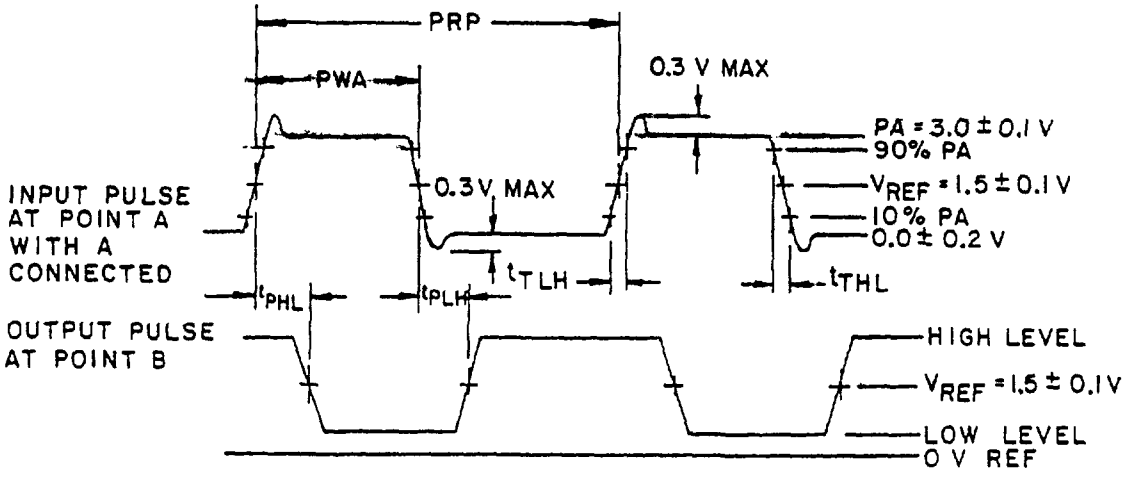
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NOTES
1. R1 = 51 ±1%, 1/4 W
2. C1-C21 = 50 ±5 pF (including probe and fixture capacitance)
3. For test conditions Q1-Q21 and R1-R21, ground all unused input pins except for the input pin connected in column A of table XIV (1A2, 1A3, 1A4, 1A9, 1A12, 1A13, 1A16, 1A17, 1A19, 2A2, 2A3, 2A4, 2A5, 2A6, 2A7, 2A8, 2A9, 2A13, 2A15, and 2A16)

FIGURE 35 Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV

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NOTES

1. Input pulse parameters
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XIV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins, between probes YO1-1 and YO1-2 and output pins, and between load circuits and output pins, shall be less than 2 inches (50.8 mm).
4. All voltages are referenced to ground

FIGURE 36 Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV

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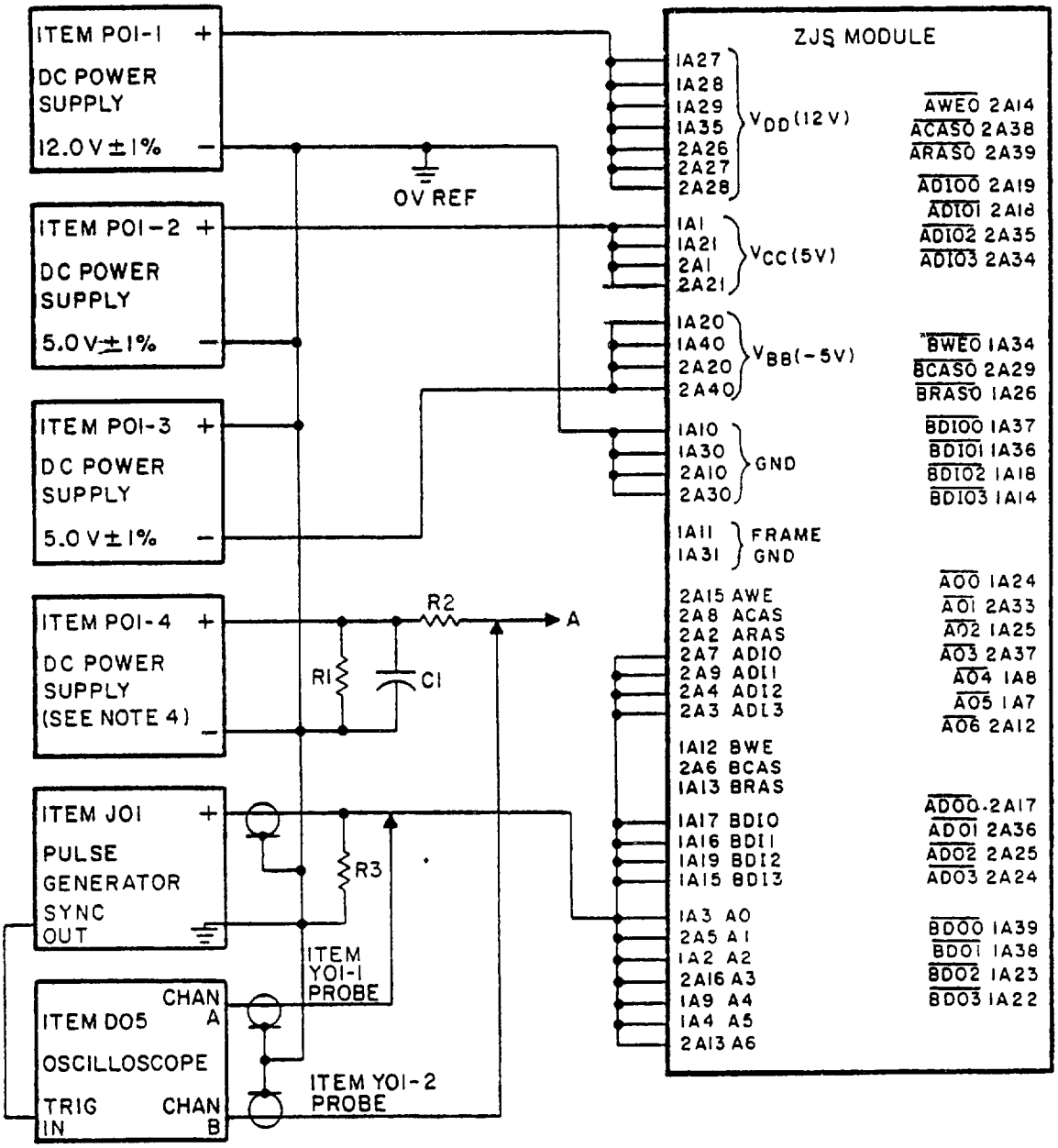
TABLE XV Input crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
S1	AWE	Input crosstalk, low level	2A15	0	500	mV	37,38
S2	ACAS		2A8				
S3	ARAS		2A2				
S4	BWE		1A12				
S5	BCAS		2A6				
S6	BRAS		1A13				
T1	AWE	Input crosstalk, high level	2A15				
T2	ACAS		2A8				
T3	ARAS		2A2				
T4	BWE		1A12				
T5	BCAS		2A6				
T6	BRAS		1A13				

TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	V_{DD}	Power supply transient amplitude	D	Open	Open	A	0	500	mV	39,40
U2	V_{CC}	Power supply transient amplitude	Open	D	Open	B	0	500	mV	39,40
U3	V_{BB}	Power supply transient amplitude	Open	Open	D	C	0	500	mV	39,40

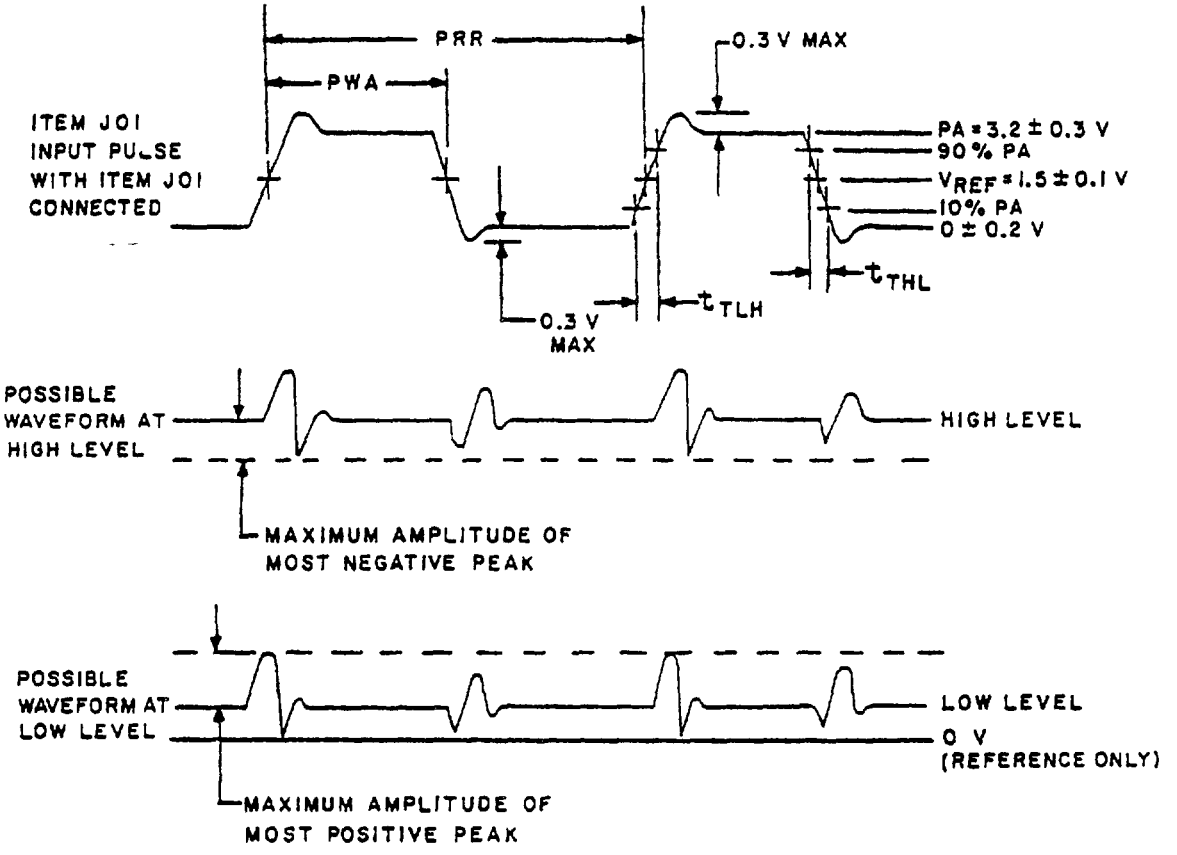
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NOTES:
1. R1 = R3 = 51Ω ±5%, 1/2 W.
2. R2 = 150Ω ±1%, 1/4 W.
3. C1 = Sufficient capacitance to reduce item P01-4 power supply noise to ≤ 50 mV.
4. For test conditions S1-S6 adjust item P01-4 to 0.8 V ±1 and for test conditions T1-T6 adjust to 2.0 V ±1%.

FIGURE 37. Input crosstalk (low-level and high-level) test circuit for table XV.

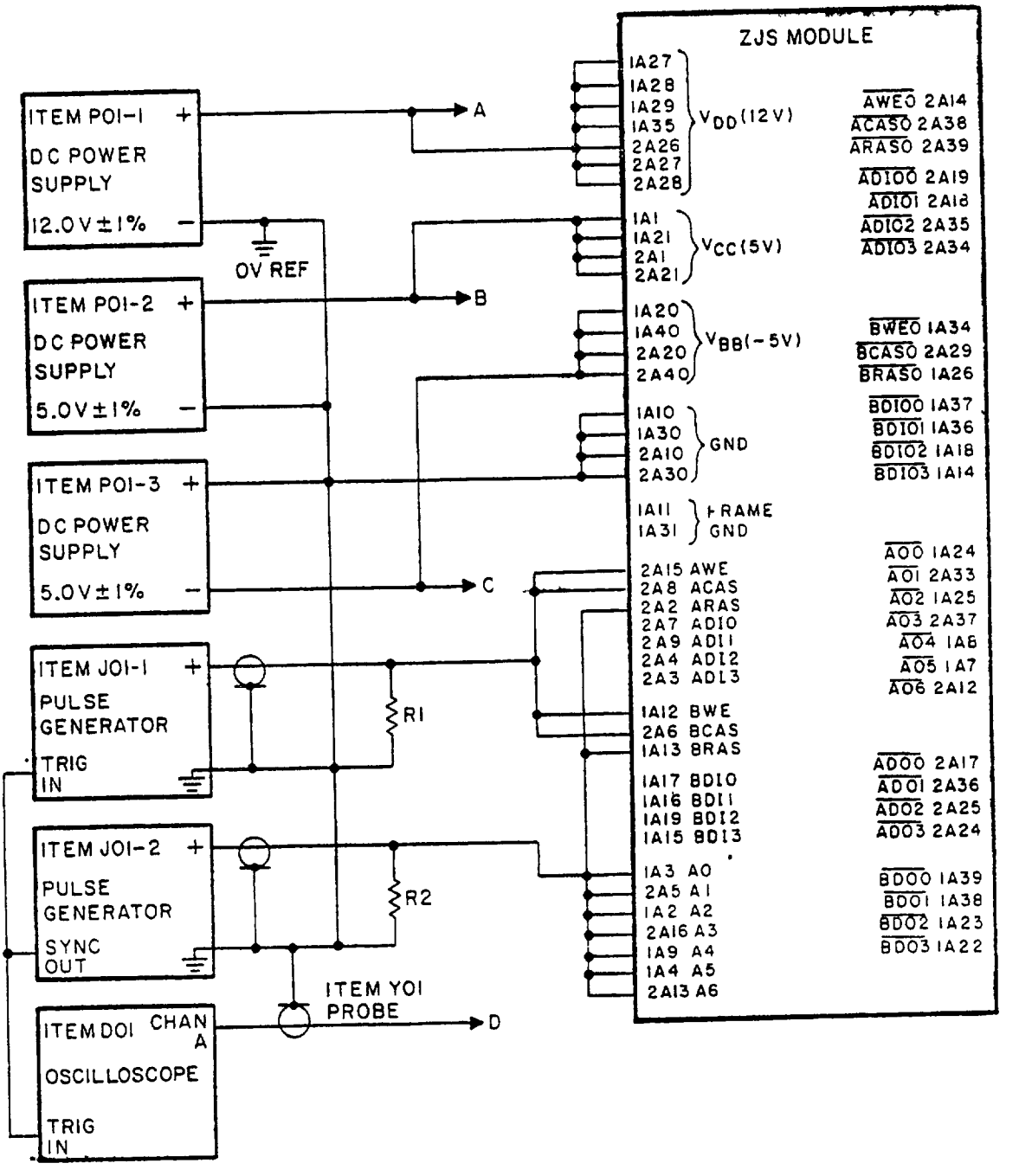
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NOTES
1 Input pulse parameters:
$t_{THL} = t_{TLH} = 20 \pm 2$ ns.
PWA = 500 ±50 ns.
PRR = 1,000 ±100 ns.
2 The lead length of electrical conductors between terminator resistor R3 and input pins, probes Y01-1, Y01-2 to input pins, and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

FIGURE 38 Input crosstalk (low-level and high-level) test circuit waveforms for table XV

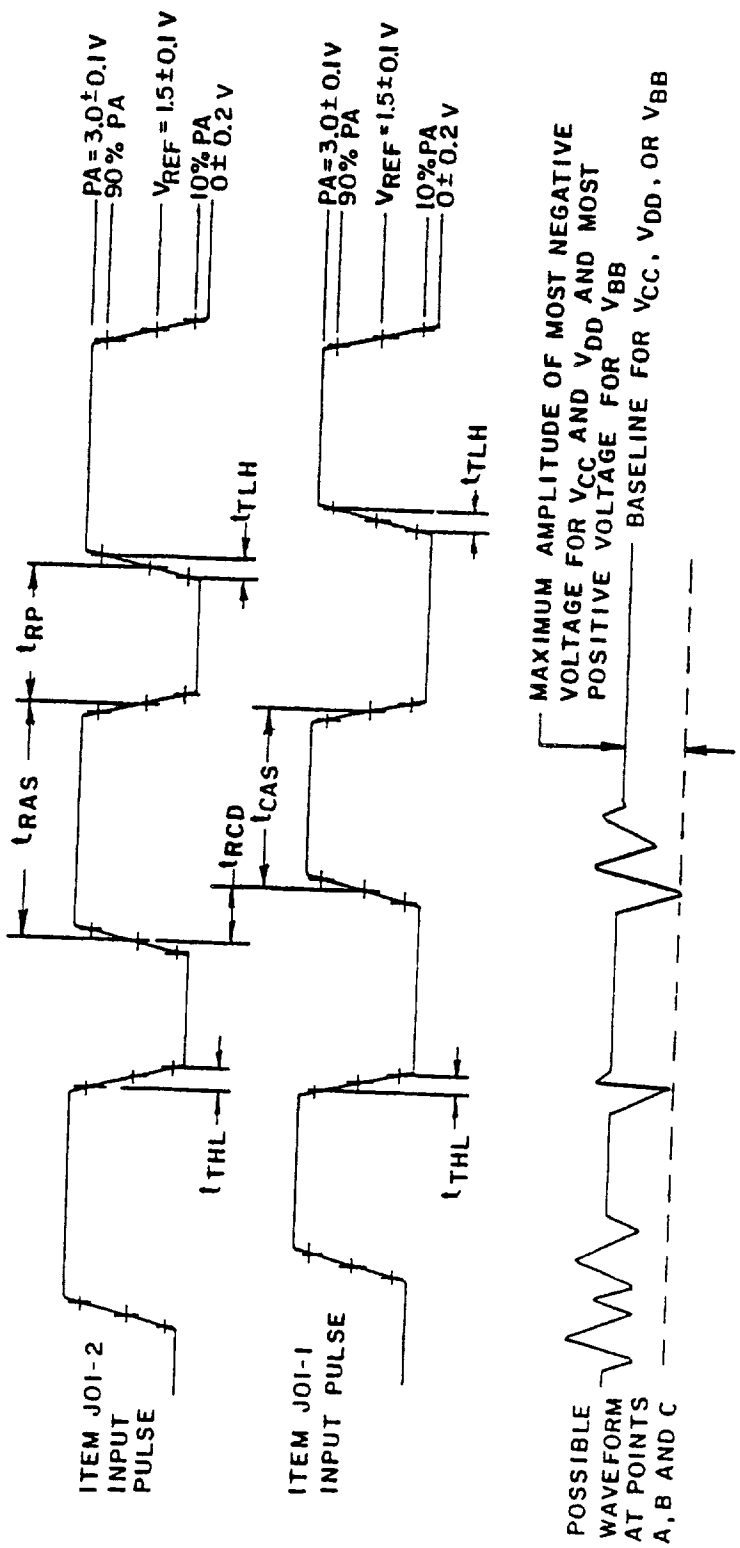
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NOTE: R1 = R2 = 51Ω ±1%, 1/4 W

FIGURE 39 Power supply transient amplitude test circuit for table VI

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NOTE Input pulse parameters
$t_{TLH} = t_{THL} = 10 \pm 2$ ns
t_{RCD} = 200 +5, -0 ns
t_{RAS} = 350 +5, -0 ns
t_{CAS} = 300 +5, -0 ns
t_{RP} = 150 +5, -0 ns.

FIGURE 40 Power supply transient amplitude test circuit waveforms for table XVI.

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TABLE XVII Power supply filter, input, and output capacitance test.

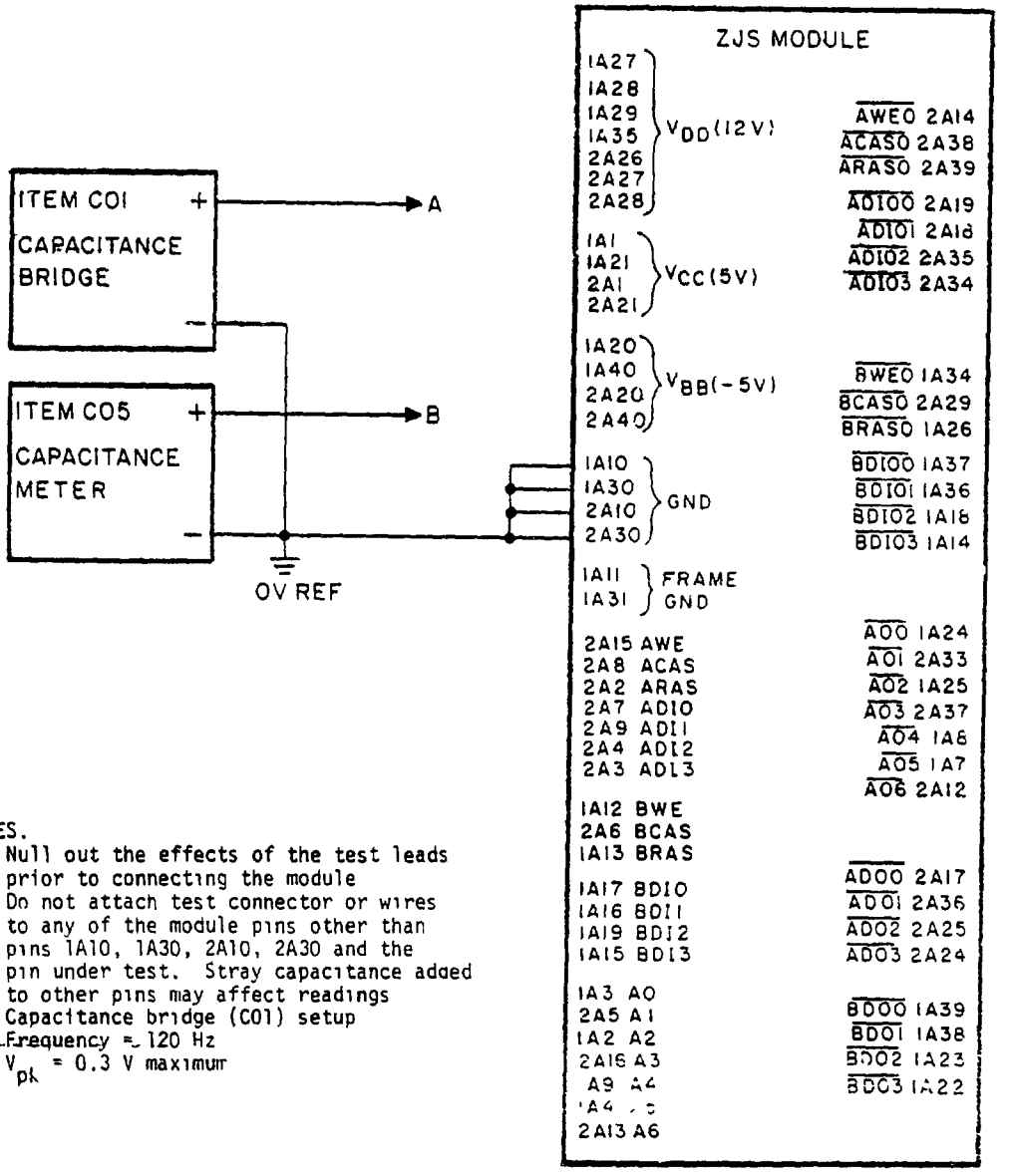
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
V1	V_{DD}	Power supply filter capacitance	1A27	Open	13.2	24.6	µF	41
V2	V_{CC}	Power supply filter capacitance	1A1	Open	1.54	2.86	µF	
V3	V_{BB}	Power supply filter capacitance	1A20	Open	3.12	5.78	µF	
W1	AWE	Input capacitance	Open	2A15	2	15	pF	
W2	ACAS			2A8				
W3	ARAS			2A2				
W4	ADIO			2A7				
W5	ADI1			2A9				
W6	ADI2			2A4				
W7	ADI3			2A3				
W8	BWE			1A12				
W9	BCAS			2A6				
W10	BRAS			1A13				
W11	BDIO			1A17				
W12	BDI1			1A16				
W13	BDI2			1A19				
W14	BDI3			1A15				
W15	A0			1A3				
W16	A1			2A5				
W17	A2			1A2				
W18	A3			2A16				
W19	A4			1A9				
W20	A5			1A4				
W21	A6			2A13				

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TABLE XVII. Power supply filter, input, and output capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
X1	$\overline{ADO0}$	Output capacitance	Open	2A17	2	15	pF	41
X2	$\overline{ADO1}$			2A36				
X3	$\overline{ADO2}$			2A25				
X4	$\overline{ADO3}$			2A24				
X5	$\overline{BDO0}$			1A39				
X6	$\overline{BDO1}$			1A38				
X7	$\overline{BDO2}$			1A23				
X8	$\overline{BDO3}$			1A22				

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NOTES.
1. Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or wires to any of the module pins other than pins 1A10, 1A30, 2A10, 2A30 and the pin under test. Stray capacitance added to other pins may affect readings
3 Capacitance bridge (CO1) setup
Frequency = 120 Hz
V_{pk} = 0.3 V maximum

FIGURE 41 Power supply, data, input and output capacitance test circuit for tab e ...

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